



F107 – Foundations of Computer Science

Lecture 5

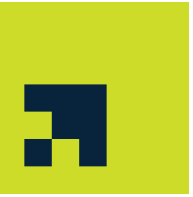
Dr. Sanjit Kumar Saha

Topics

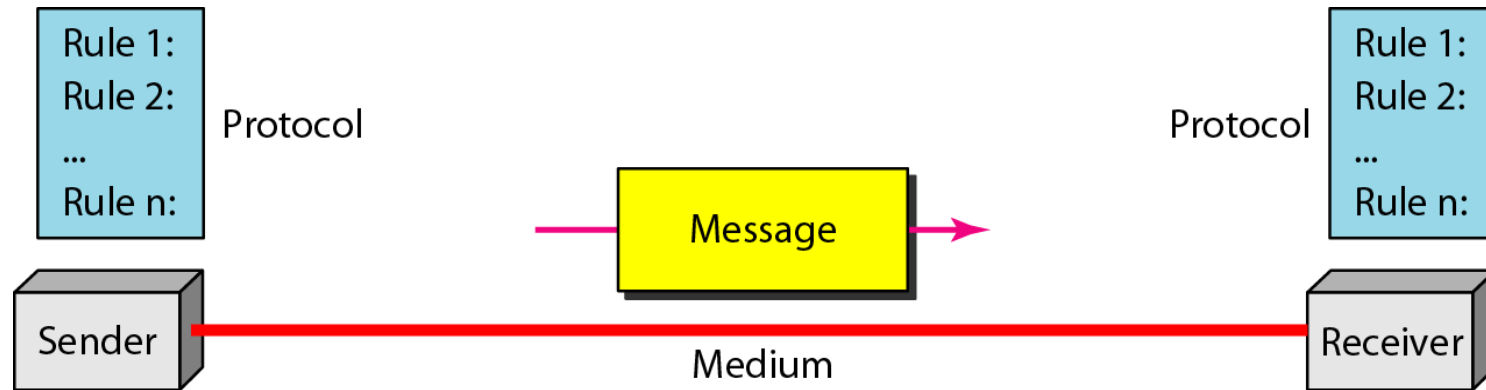


- Computer Networks
 - Computer networks and topologies
 - Protocols
 - TCP/IP Layers
- Corresponds to Book Chapter 6

Computer Networks



- a set of devices (often referred to as nodes) connected by communication links
 - a node can be a computer, printer, or any other device capable of sending and/or receiving data generated by other nodes on the network



Communication Devices: Examples



- Common communication devices



- Non-common communication devices

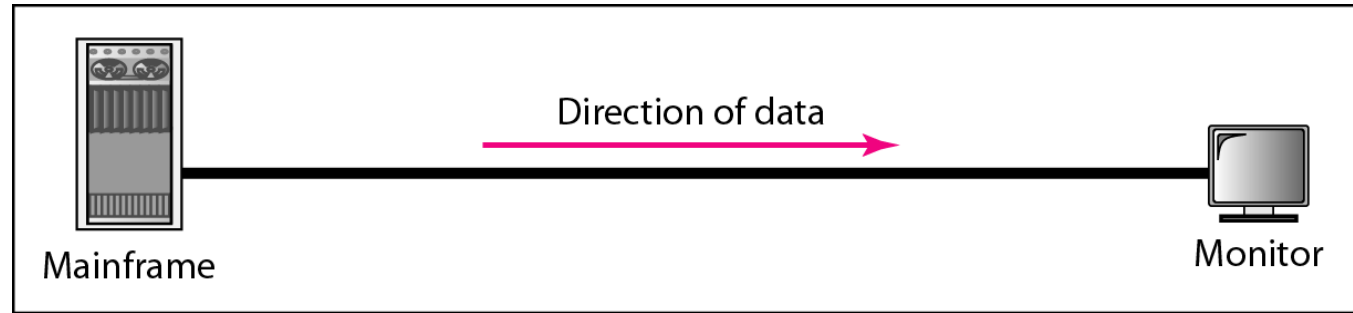
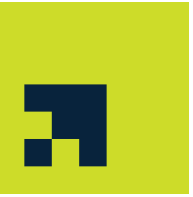


- Bluetooth watches

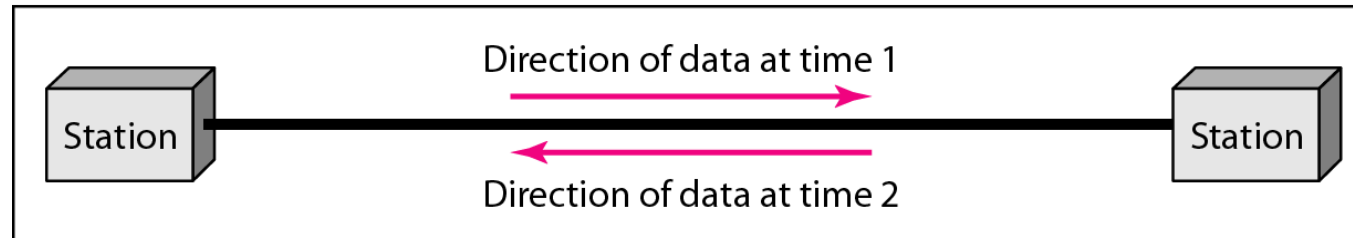


- Bluetooth alarm clock

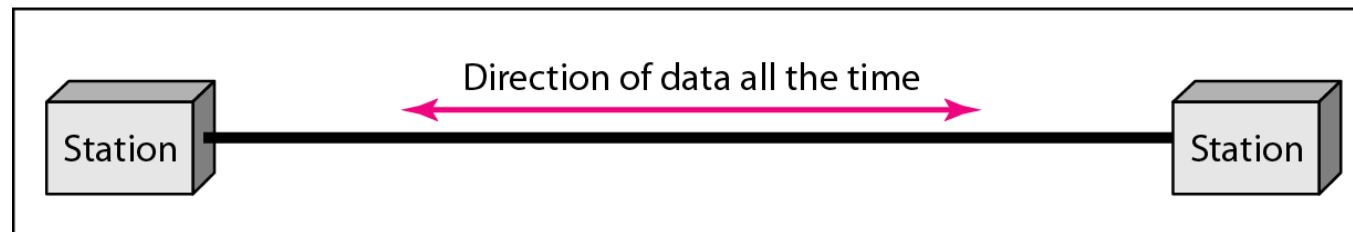
Data flow



a. Simplex

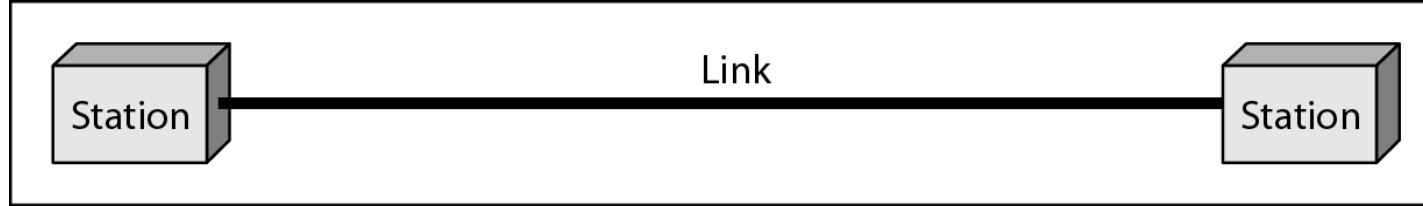
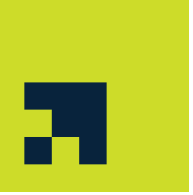


b. Half-duplex

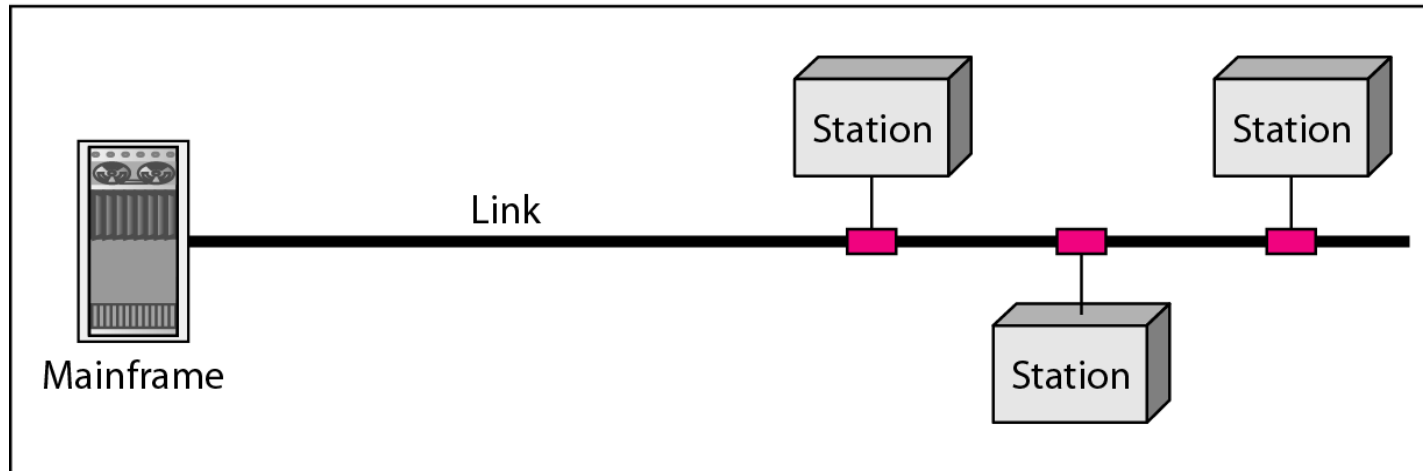


c. Full-duplex

Types of connections

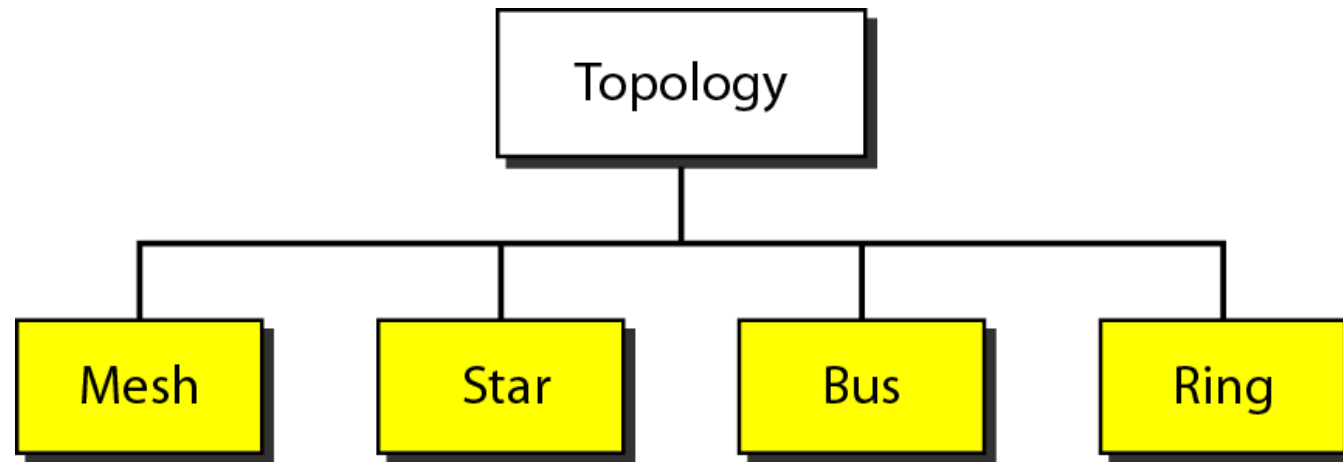
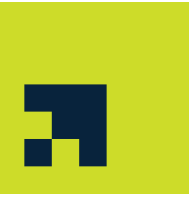


a. Point-to-point

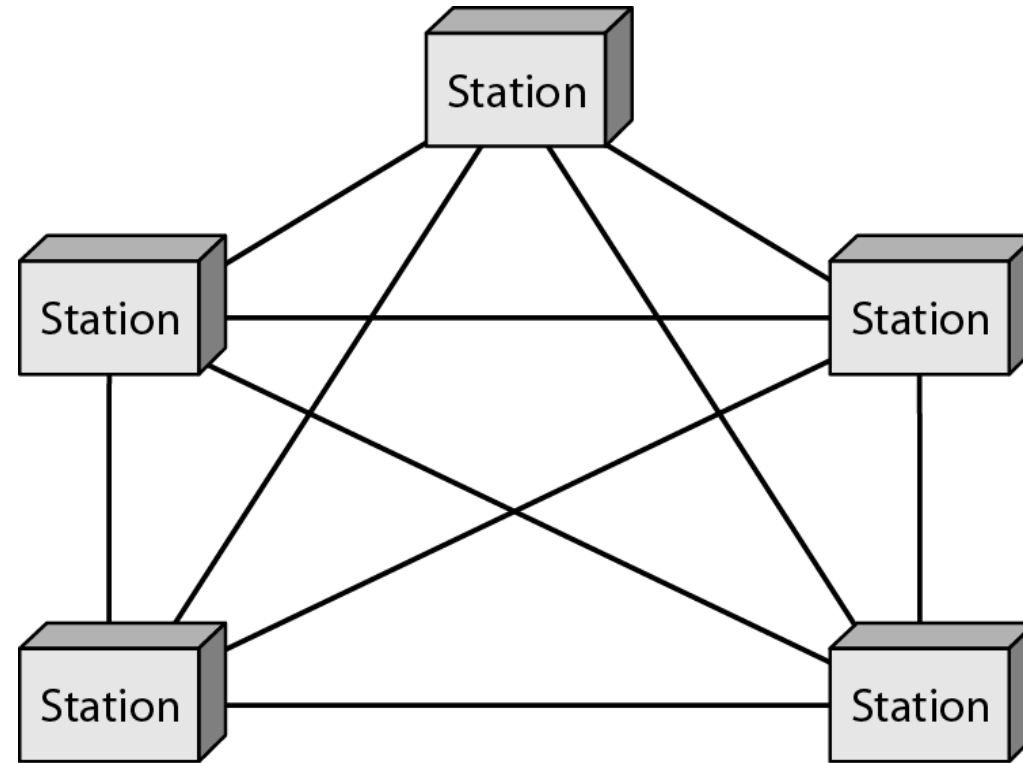
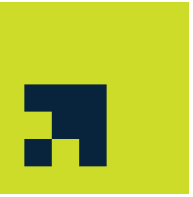


b. Multipoint

Categories of topology



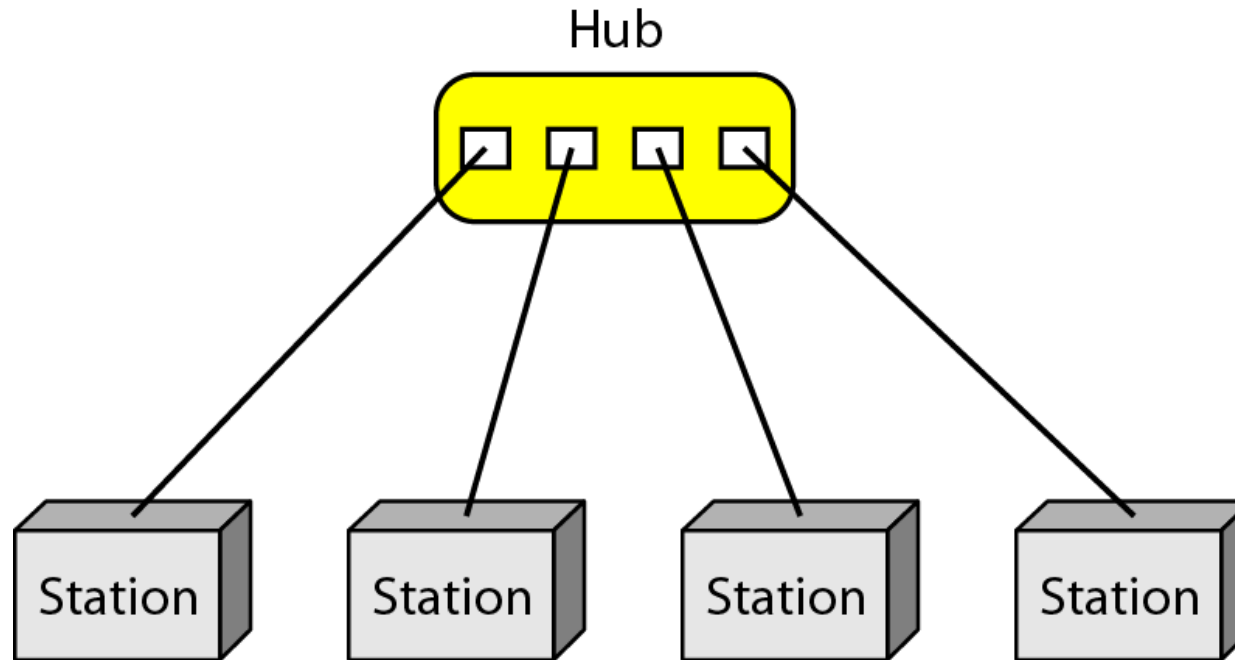
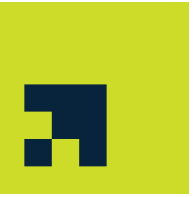
Mesh



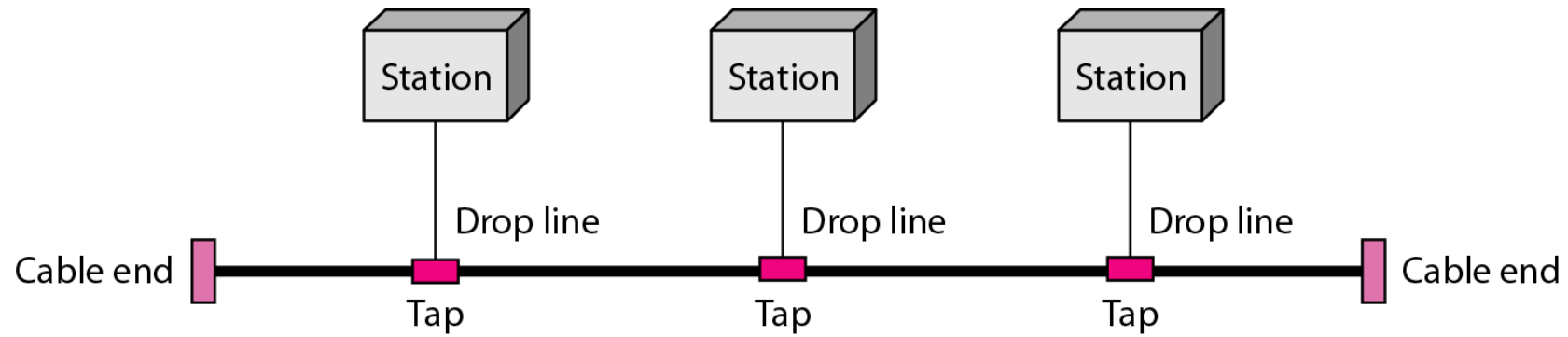
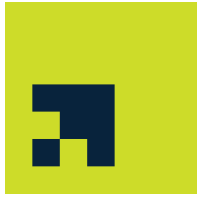
$n(n-1)$ half-duplex links

$n(n-1)/2$ duplex links

Star

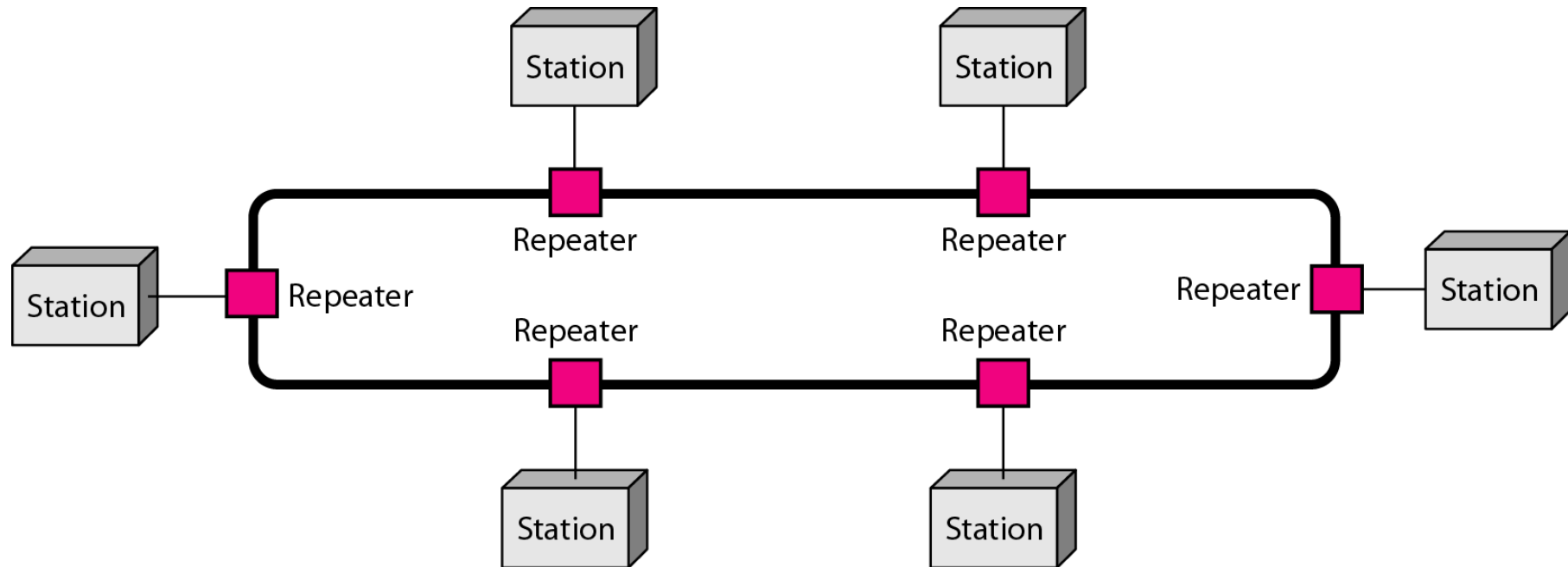
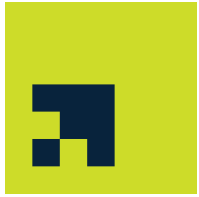


Bus



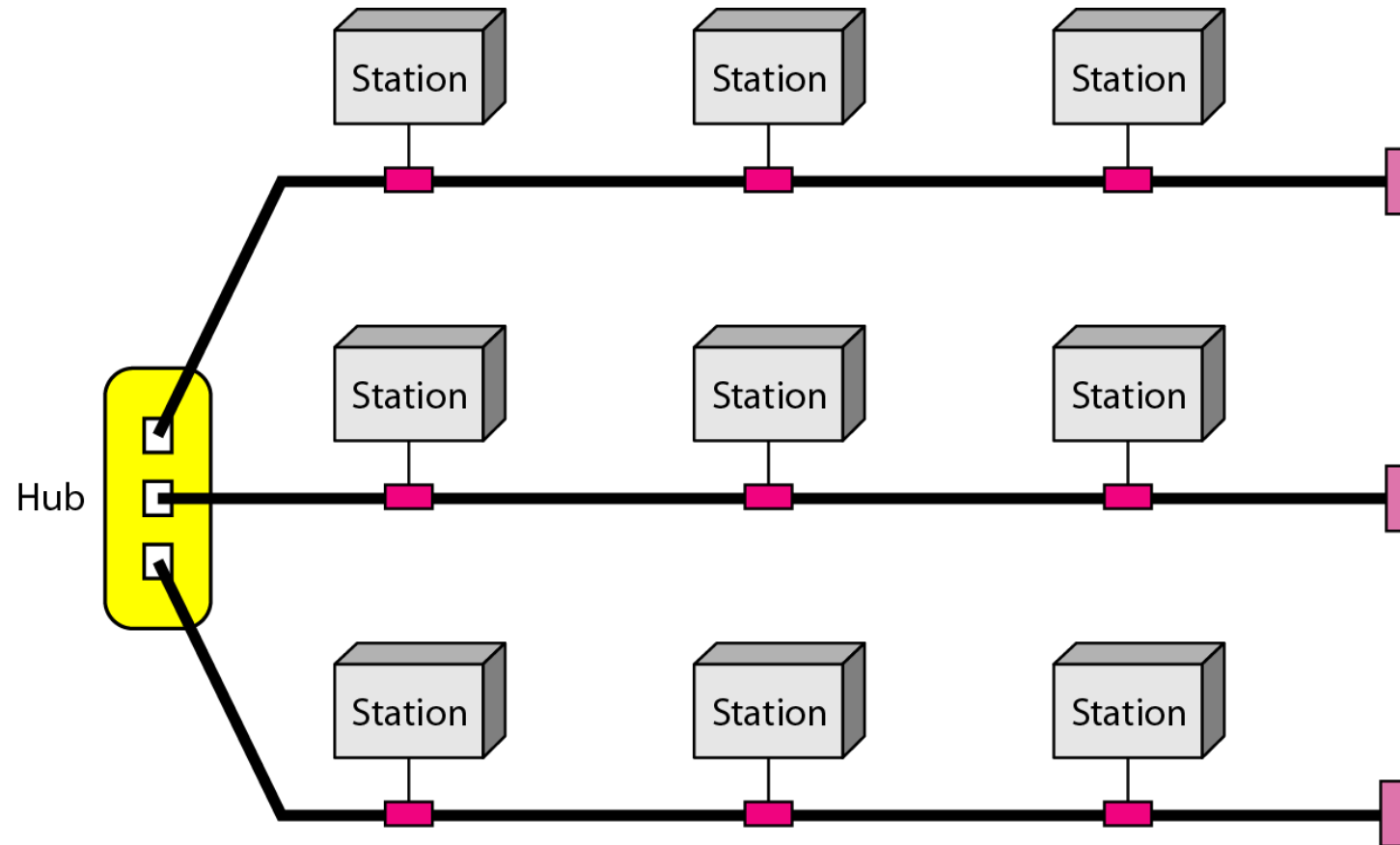
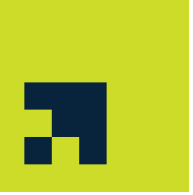
Pictures: Wikipedia

Ring



Pictures: Wikipedia

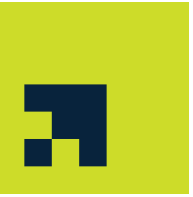
A hybrid topology



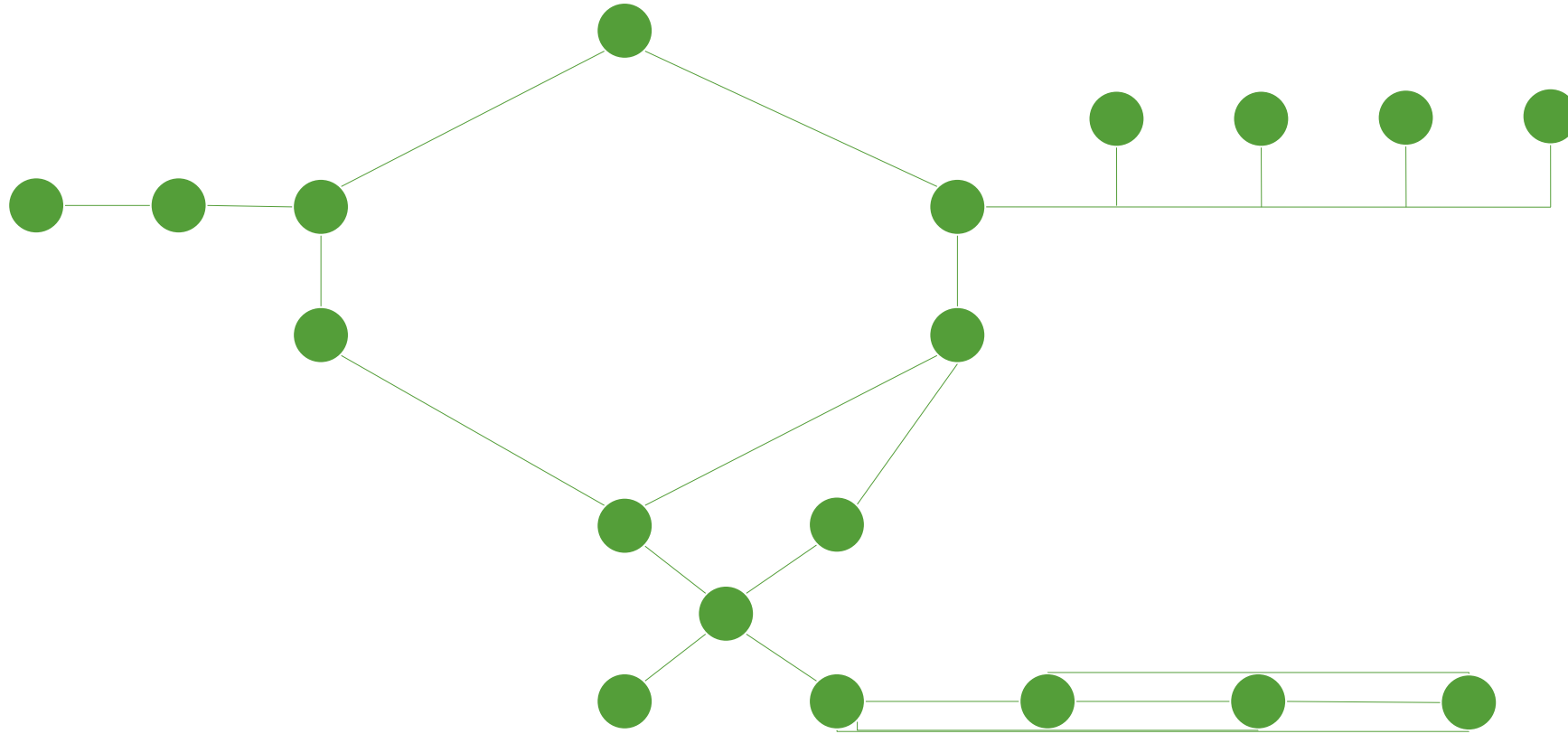
a star backbone with three bus networks

Pictures: Wikipedia

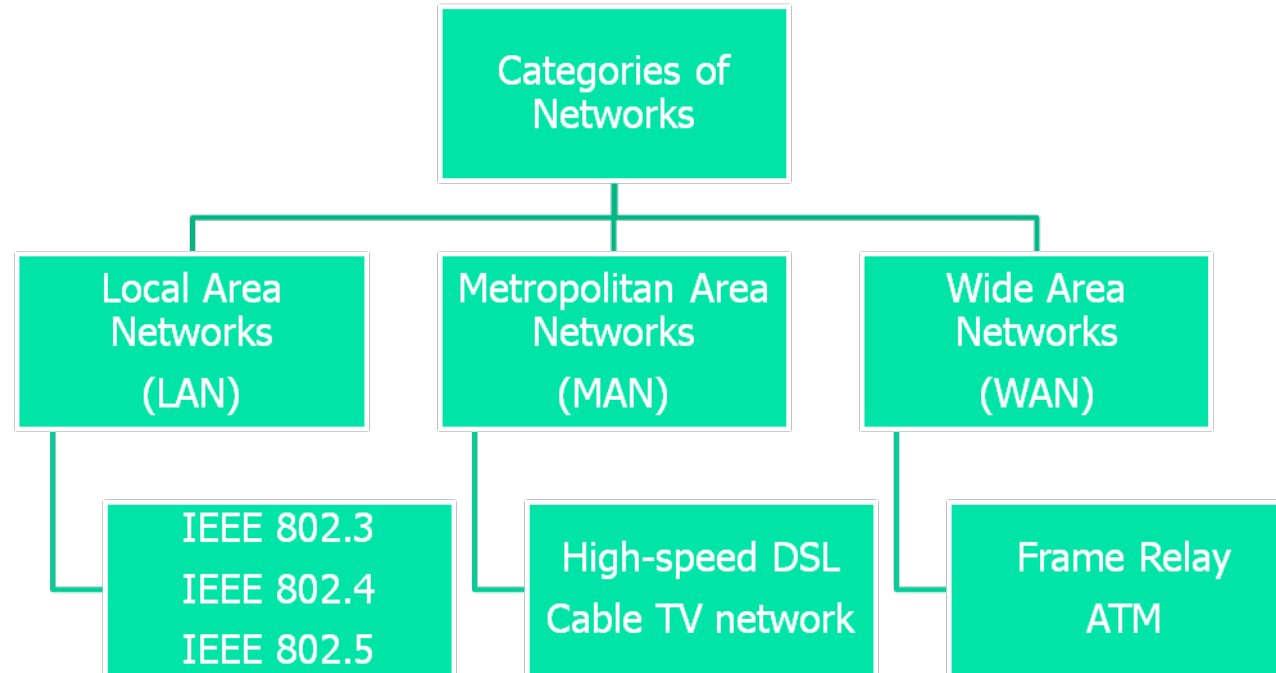
Activity



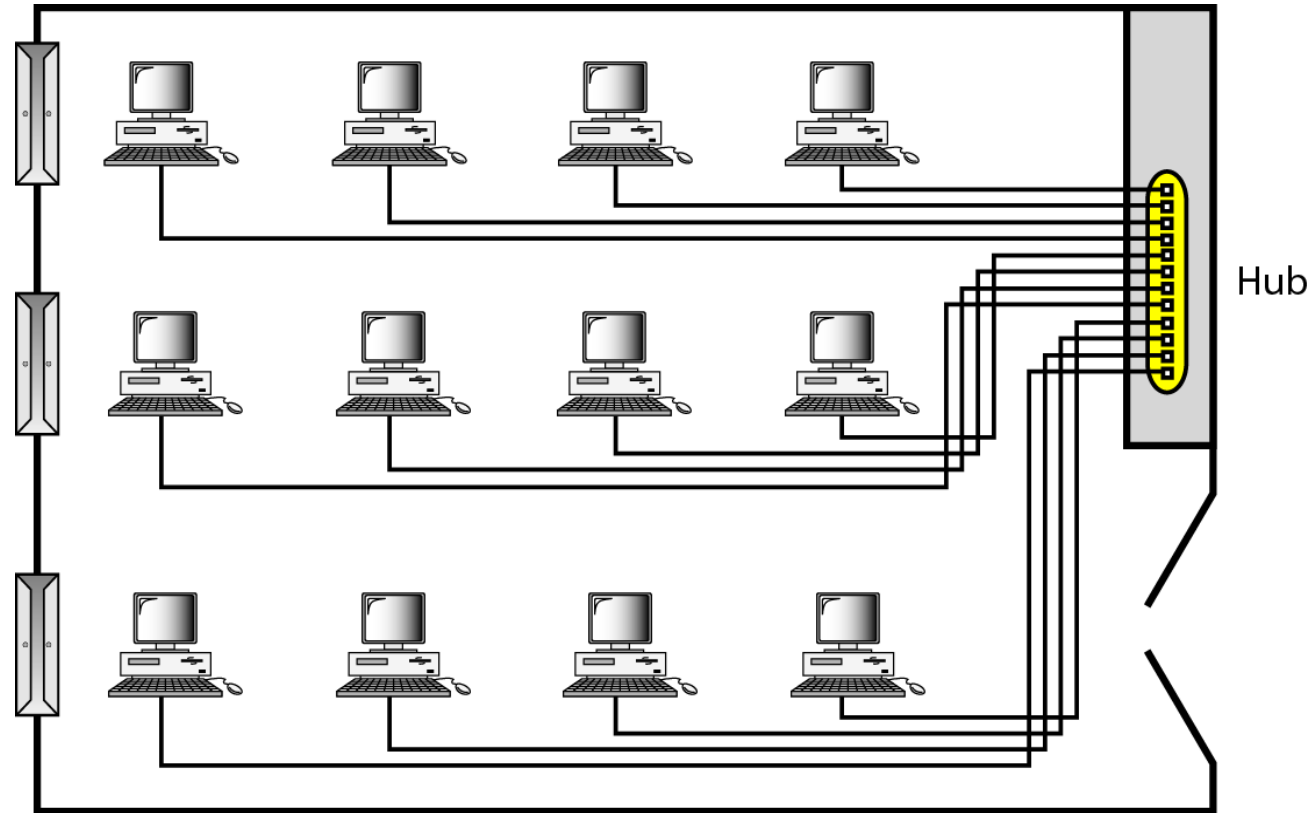
- Which topologies are used in this example?



Categories of Networks

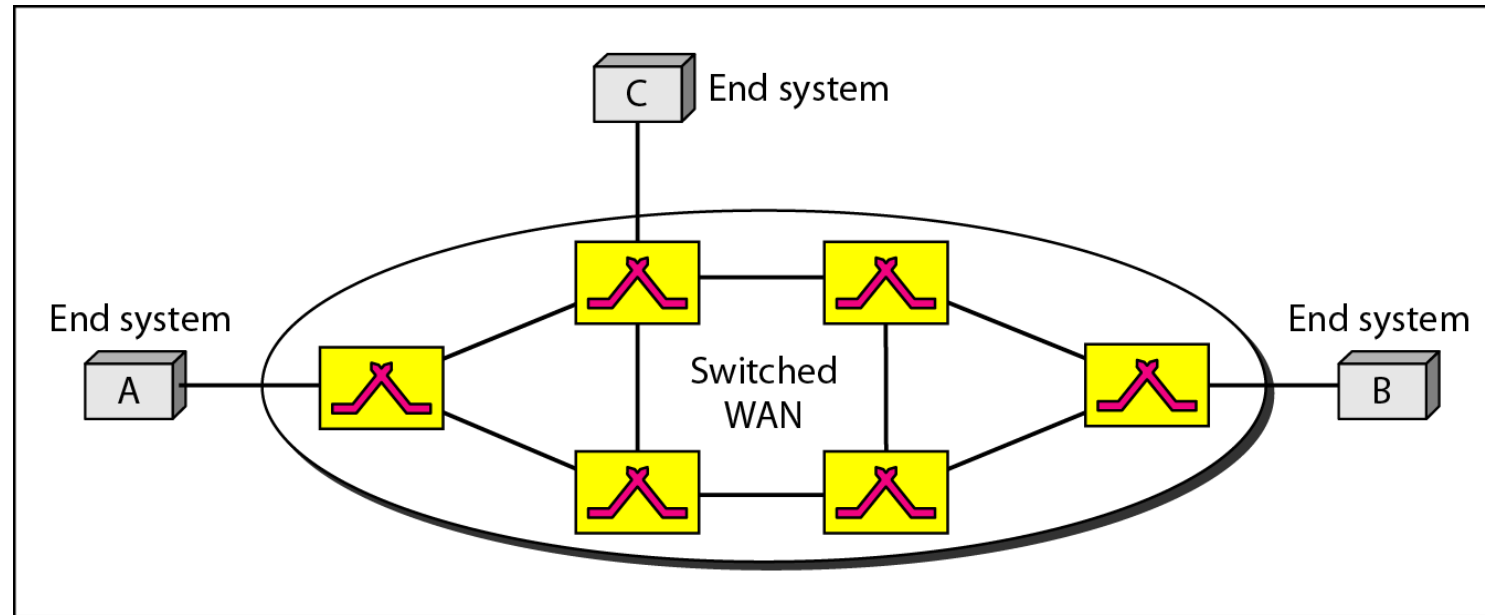
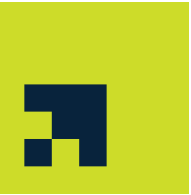


LAN

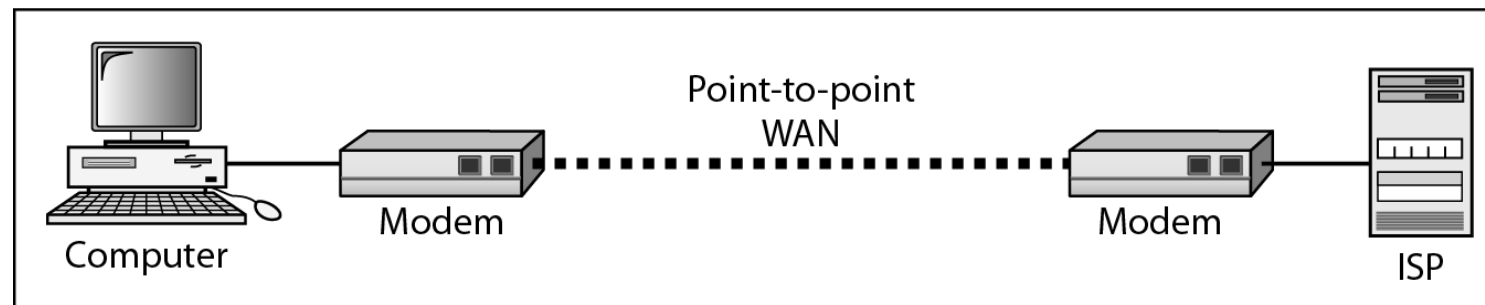


An isolated LAN connecting 12 computers to a hub in a closet

WAN

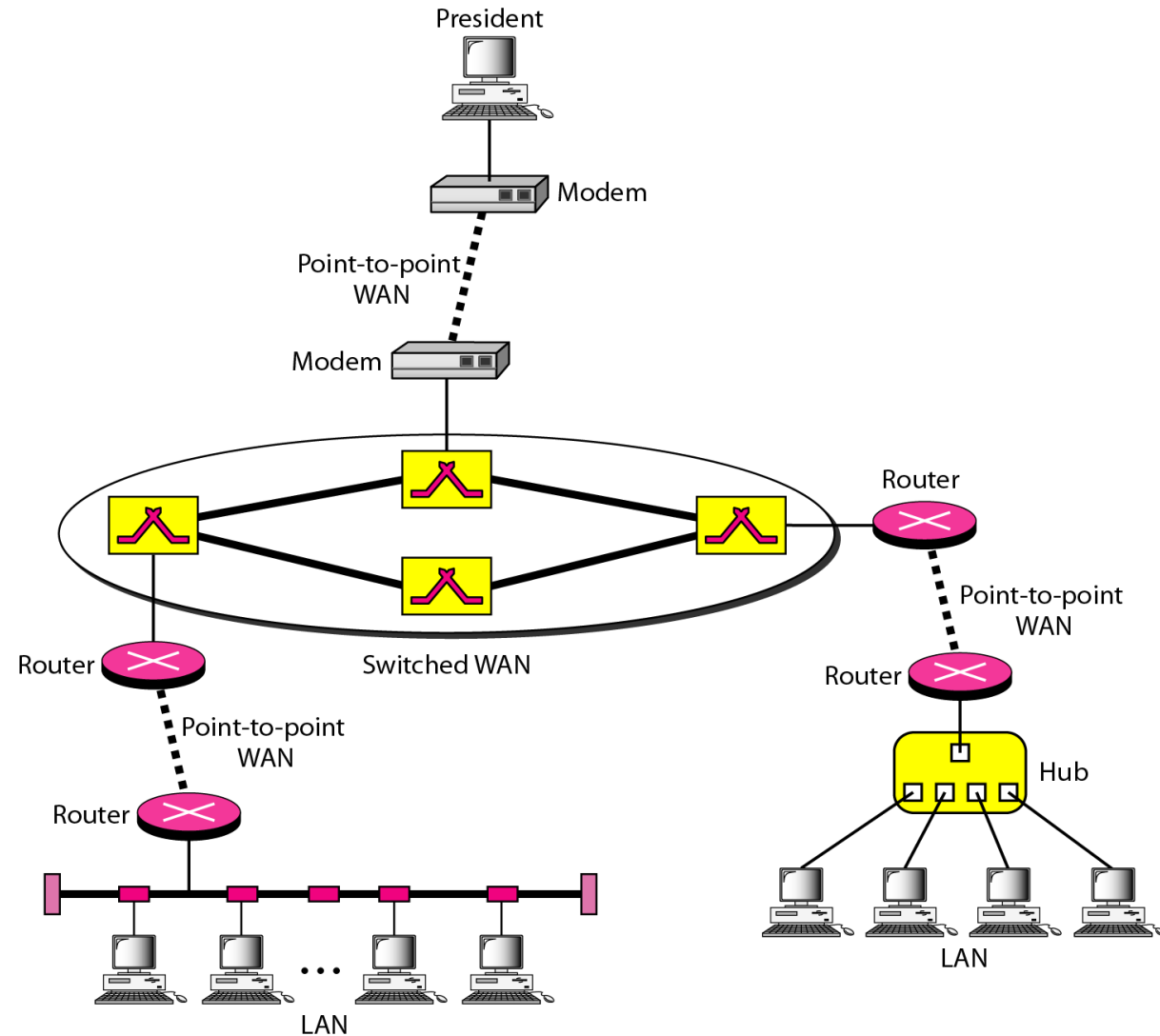


a. Switched WAN

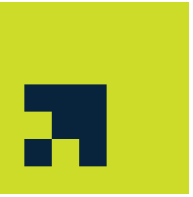


b. Point-to-point WAN

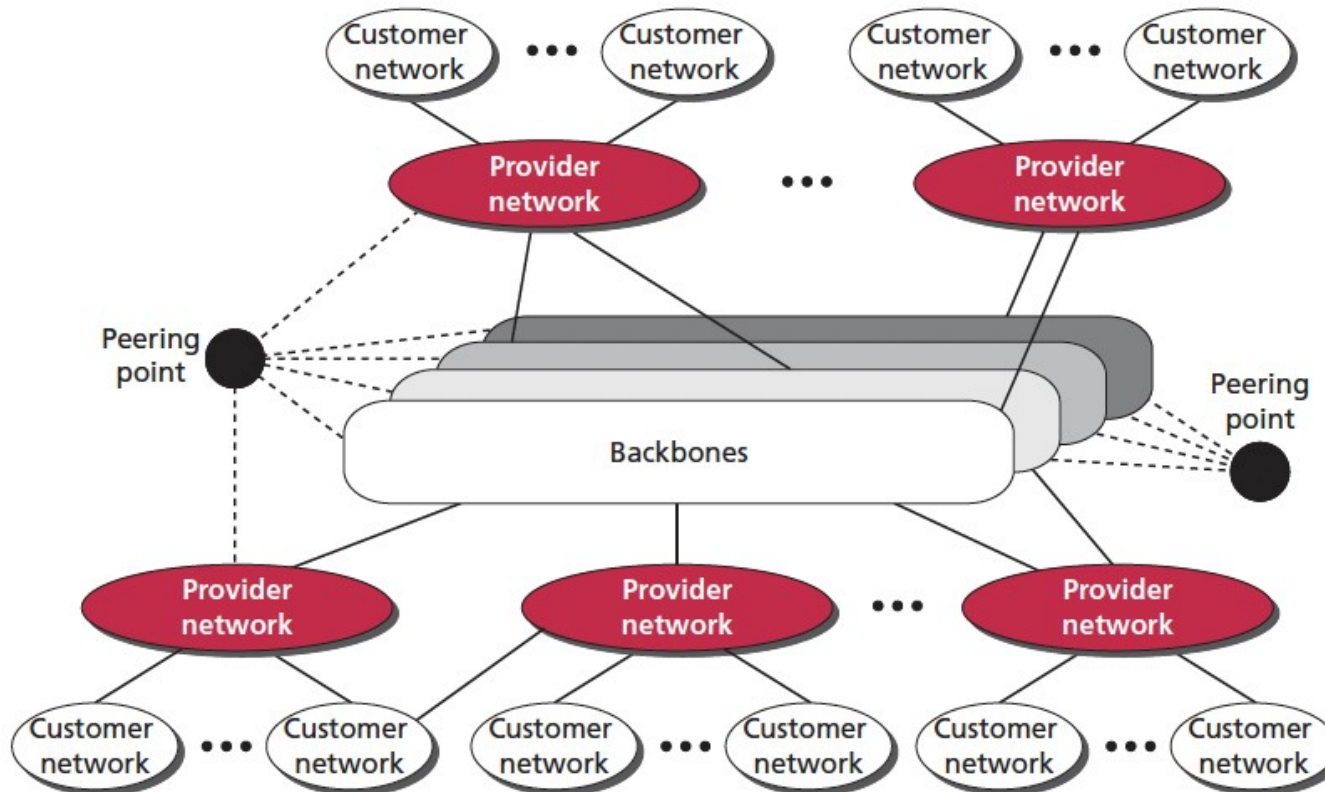
A heterogeneous network



The Internet



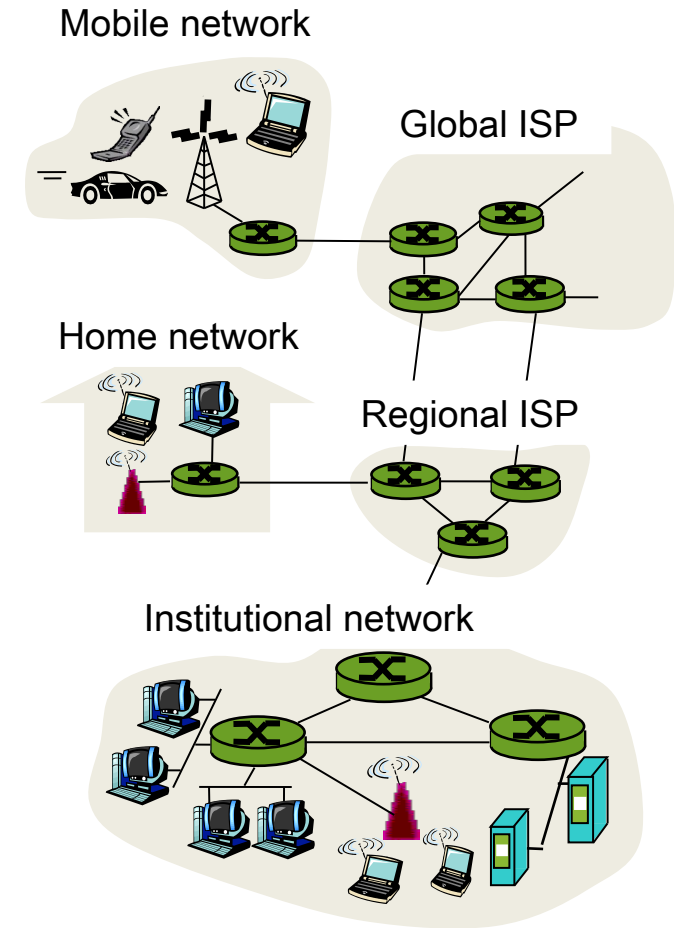
- An **internet** (note the lowercase i) is two or more networks that can communicate with each other.
- The **Internet** (uppercase I), and is composed of thousands of interconnected networks.



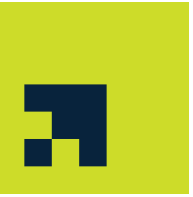
What is the Internet?

- Network of networks
 - Billions of devices
 - Independent network operators
 - Many countries
 - Laws and regulation
 - Cost
- Properties (selection)
 - Global
 - Dynamic
 - Accessible
 - Anonymous
 - Free

→ Best-Effort-Paradigm

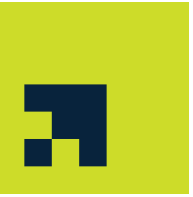


Protocol



- A protocol is a set of rules for governing data communication.
 - Defines how data is sent, received, and processed.
 - Ensures devices can communicate reliably.
 - Like a language computer use to talk to each other.

Why can a Network function?



- Many products
 - Various manufacturers, countries, markets
 - Different applications, requirements, form factors
 - Old vs. new products, different owners
- Solution: Protocol
 - Definitions of message...
 - Format
 - Order
 - Actions
 - Conditions→ Syntax and semantics
 - Provides a specific service to other entities
 - Can be exchanged and modified
 - Internal realization unimportant

Human Protocol:

A: Good morning.

B: Hello.

A: What time is it?

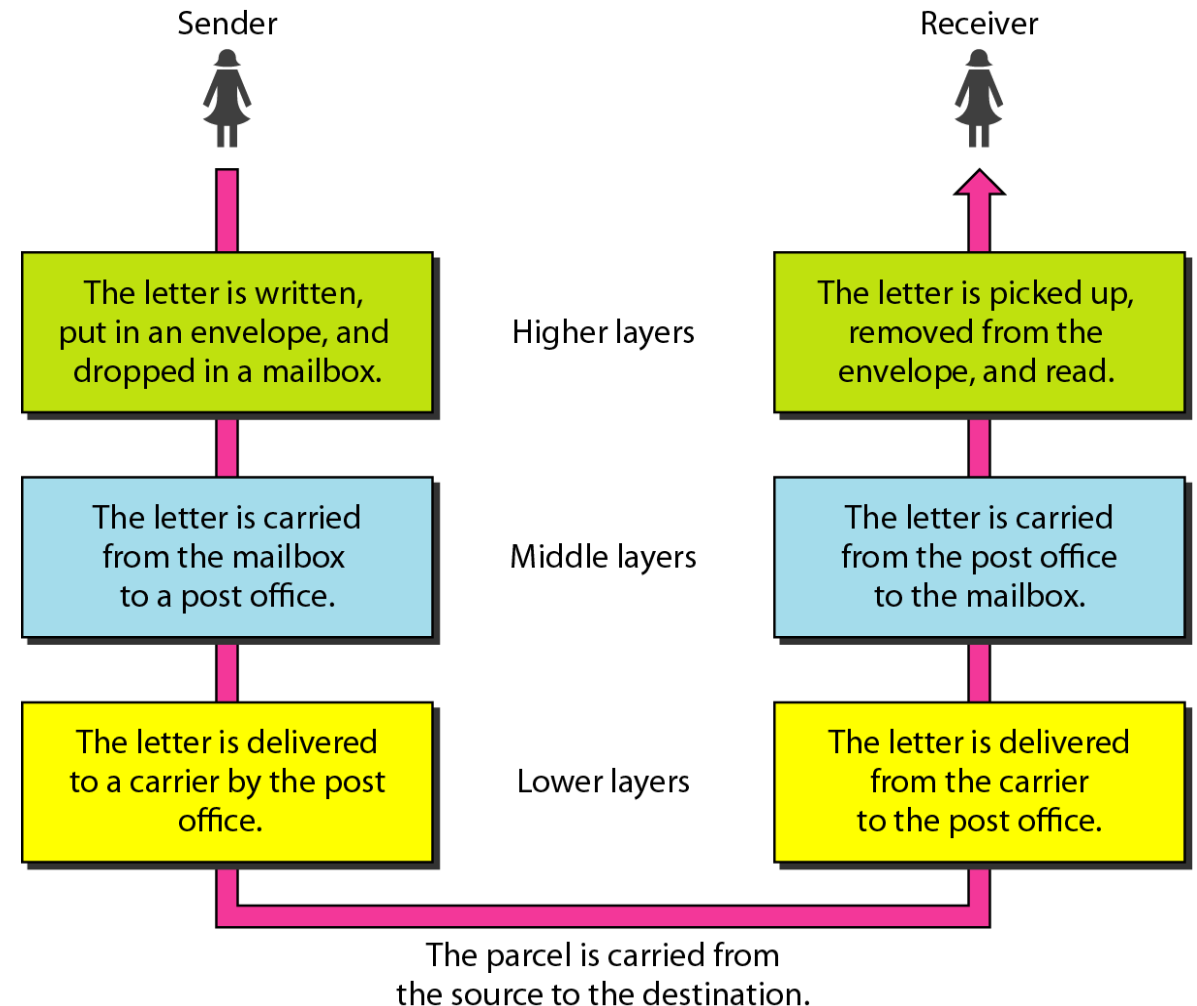
B: 13:30

A: Thanks

B: You're welcome

Layered Approach

- We use the concept of layers in our daily life.
- As an example, let us consider two friends who communicate through postal mail.
- The process of sending a letter to a friend would be complex if there were no services available from the post office.

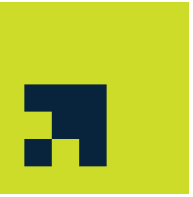


Scenarios

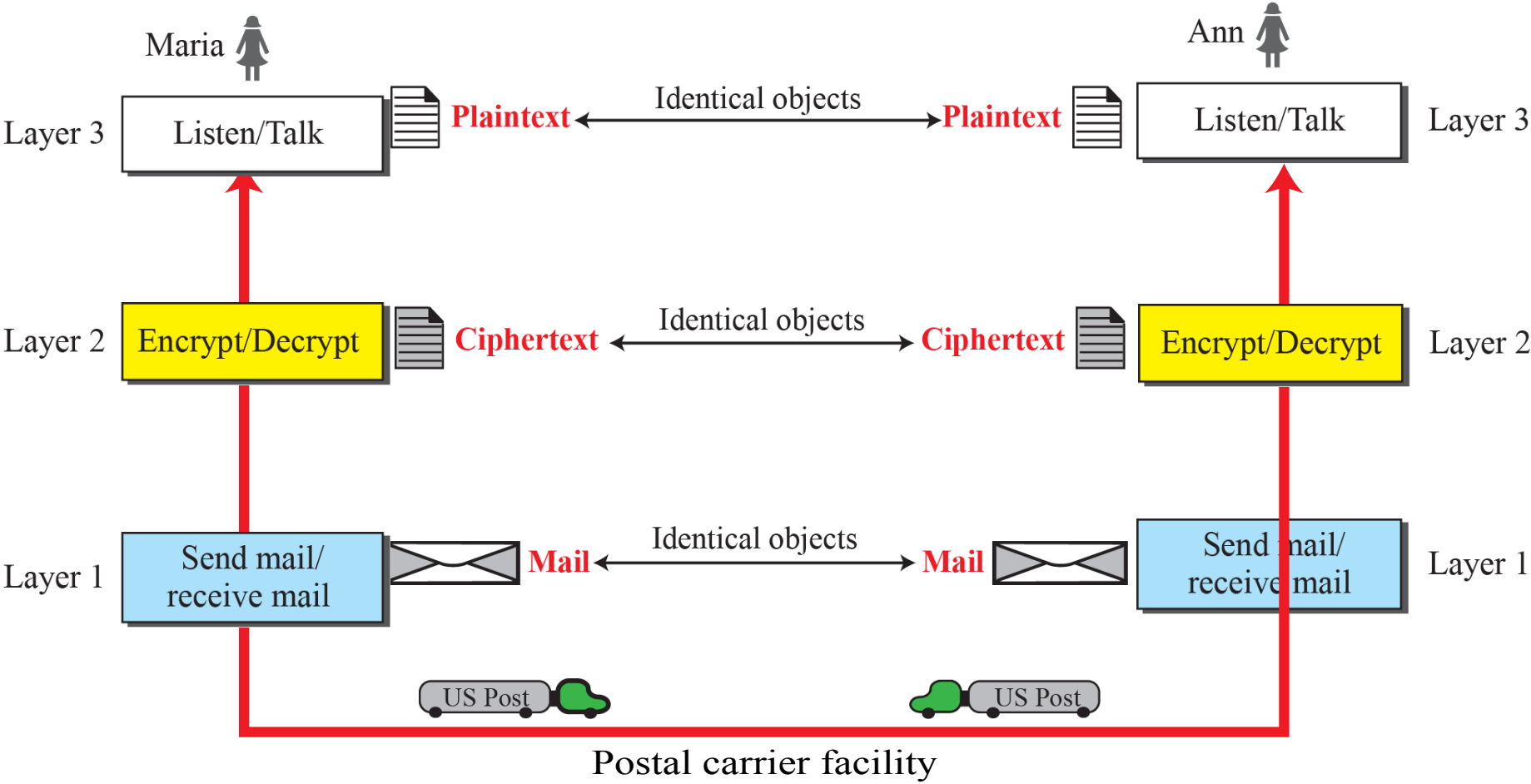
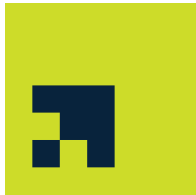


- Let us develop two simple scenarios to better understand the need for protocol layering.
- In the first scenario, communication is so simple that it can occur in only one layer.
- In the second, the communication between Maria and Ann takes place in three layers.

A single-layer protocol



A three-layer protocol

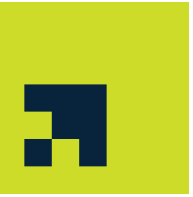


A Layered Approach – TCP/IP Protocol Suite

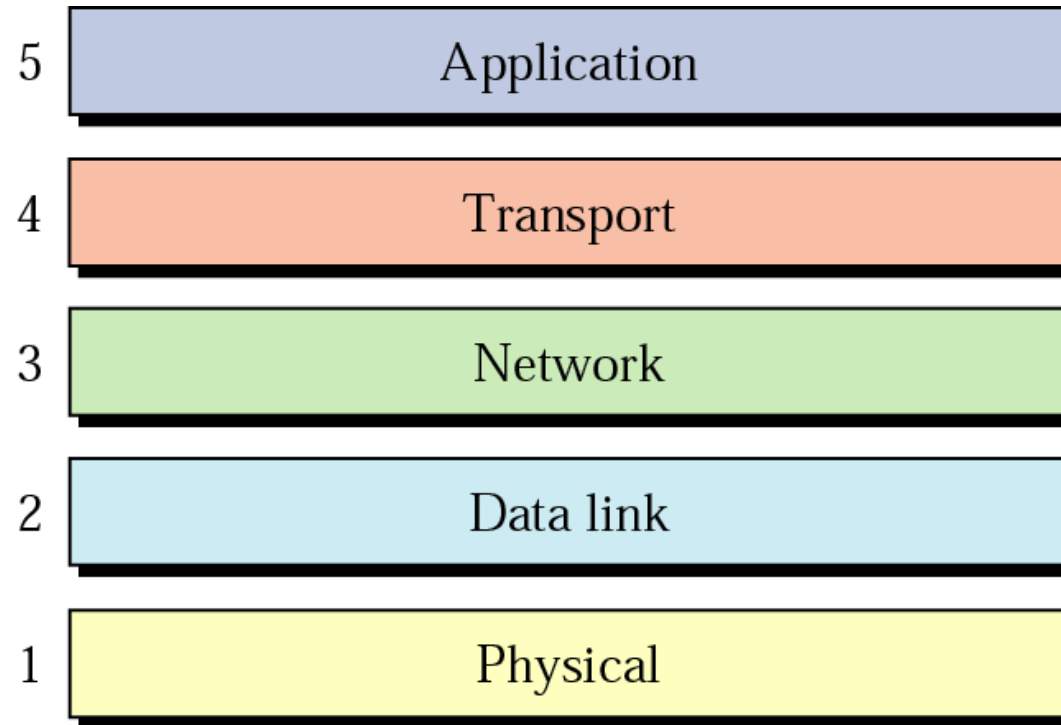


- TCP/IP is a protocol suite (a set of protocols organized in different layers) used in the Internet today.
- It is a hierarchical protocol made up of interactive modules, each of which provides a specific functionality.
- The term hierarchical means that each upper-level protocol is supported by the services provided by one or more lower-level protocols.

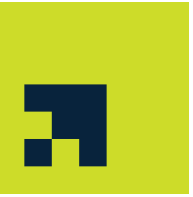
A Layered Approach – TCP/IP Protocol Suite



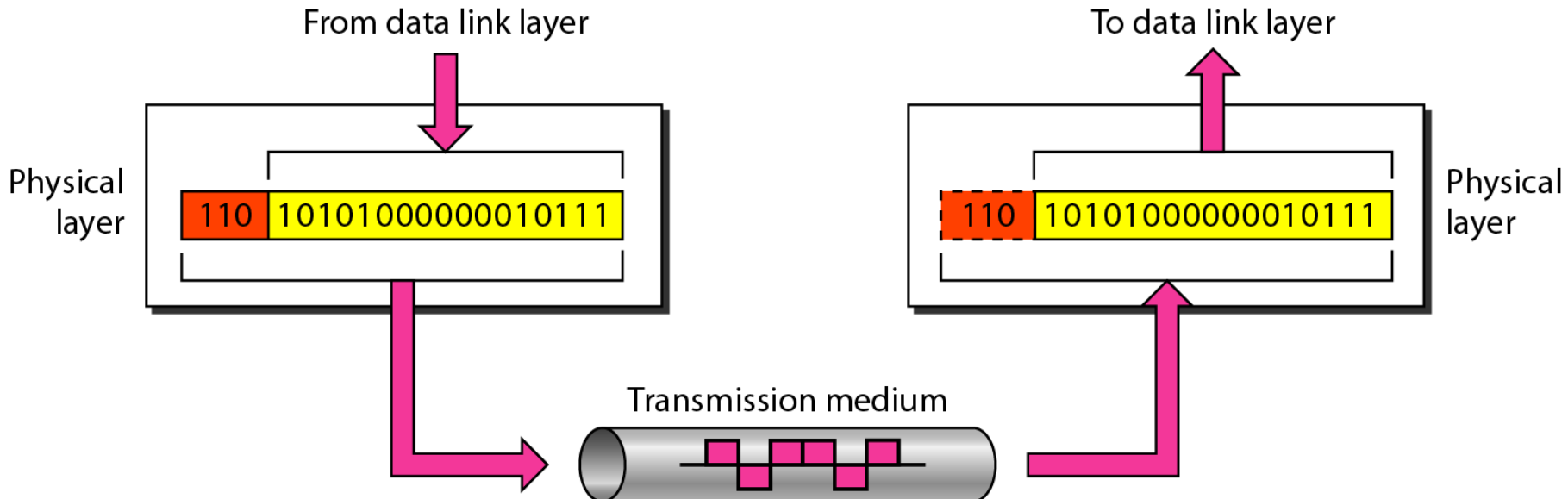
- **Layer:** A grouping of related tasks involving the transfer of information .
 - Each layer addresses an essential networking tasks.



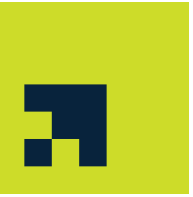
Physical layer



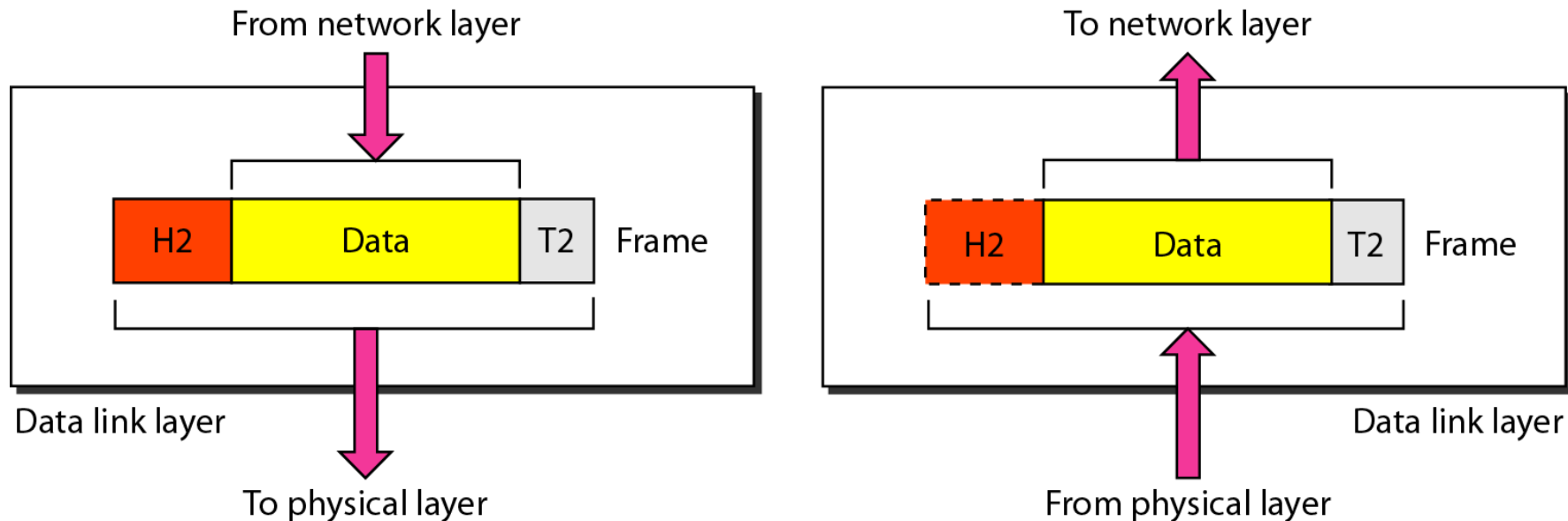
- The physical layer is responsible for movements of individual bits from one hop (node) to the next.
- Transmission Media is another hidden layer under the Physical Layer



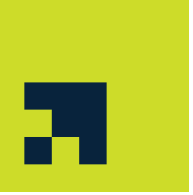
Data link layer



- The data link layer is responsible for moving frames from one hop (node) to the next over the link.
- The link can be Wired LAN/WAN or Wireless LAN/WAN.
- Some protocols provides complete error detection and correction, some protocols provide only error correction.
- At the Datalink layer addresses are called MAC (Media Access Control) Address.

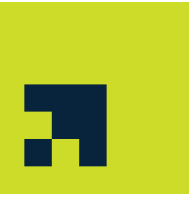


Physical address (MAC)

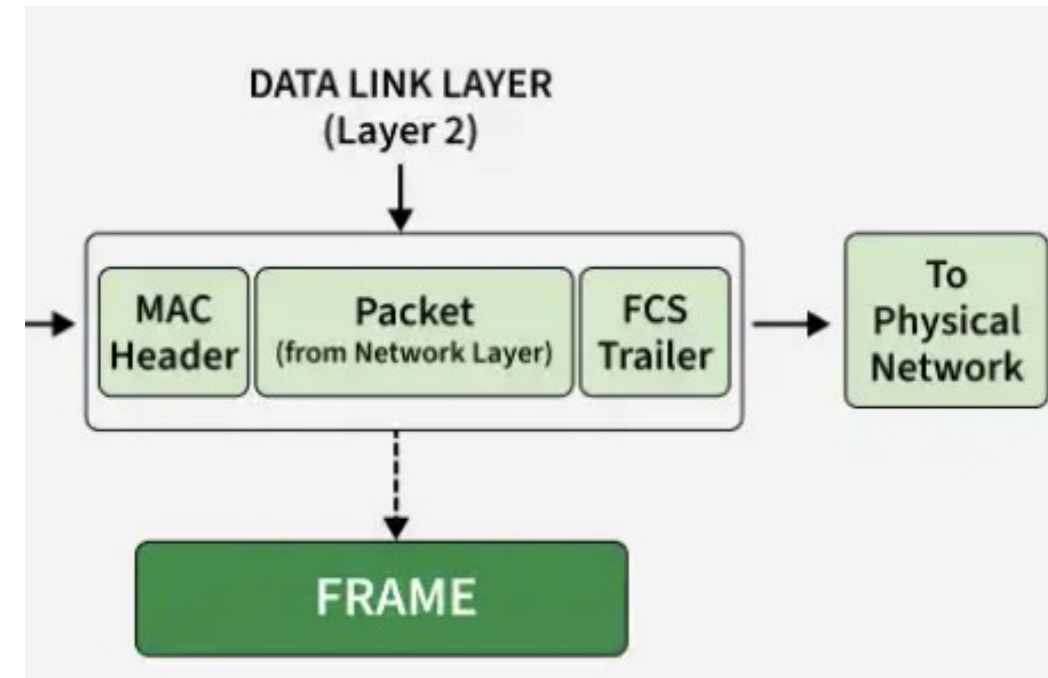


- Physical address is known also as the MAC or link address
- It is the address of a node as defined by its LAN or WAN
- It is included in the frame used by data link layer (Header)
- The MAC addresses are locally defined address

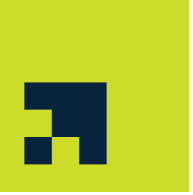
Frame



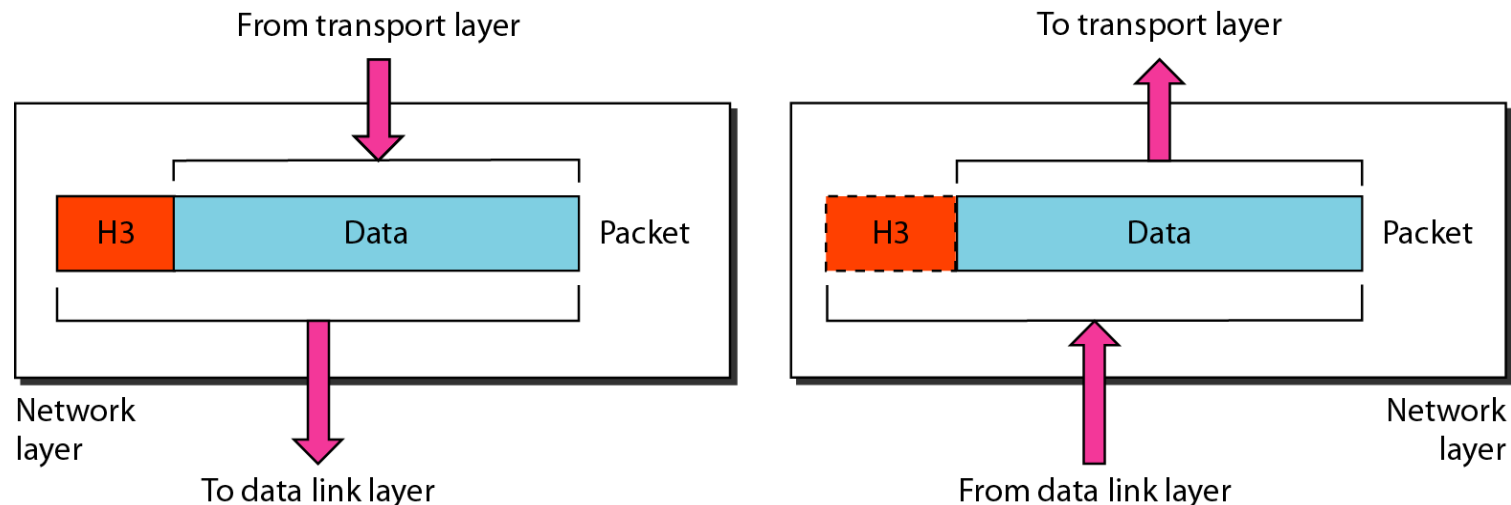
- The Key Components of a Frame are:
 - Source MAC Address: Identifies the sender's network interface.
 - Destination MAC Address: Identifies the receiver's network interface.
 - Type/Length Field: Indicates the protocol type or size of the payload.
 - Payload: Carries the IP packet from the Network Layer.
 - Frame Check Sequence (FCS): Used for error detection using CRC.



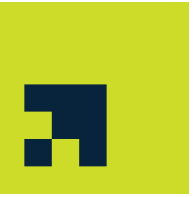
Network layer



- The network layer is responsible for the delivery of individual packets from the source host to the destination host across multiple network.
- If two system are connected to the same link (network), no need for this layer.
- The network layer also response for routing the packet.
- IP is the main protocol of the network layer.
- At the Network layer addresses are called IP Address.

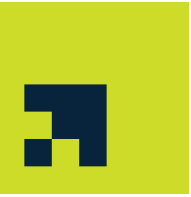


Network layer...

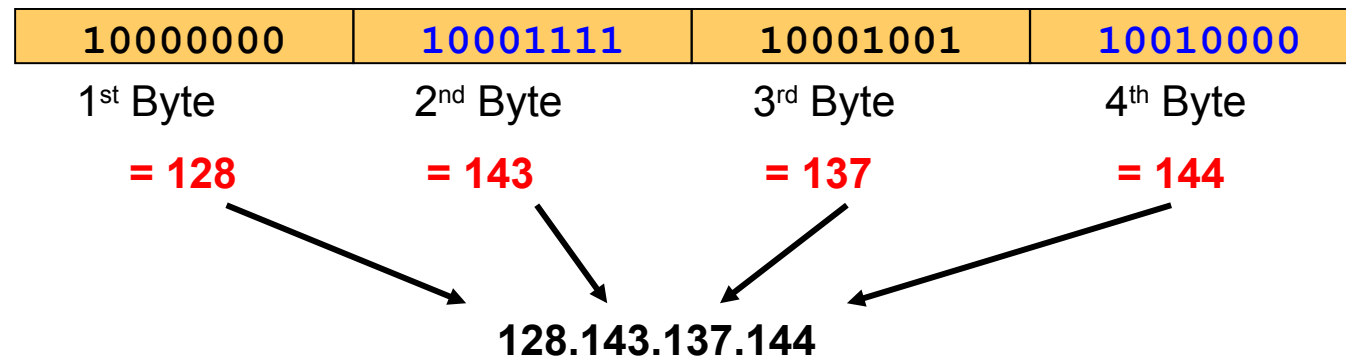


- Logical addressing (IP)
 - Network layer adds unique identifier (IP or logical address) to the packet.
 - These unique identifier(as tel. no, each tel. has unique number) enable special devices called router to make sure the packet get to correct system.
 - IP addresses are necessary for universal communications that are independent of physical network.
- Routing:
 - provide the routing mechanism for the router which route the packet to their final destination.
 - Routers : devices used when independent networks are connected to create an internetworking (network of networks)

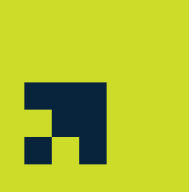
Protocol and Addressing



- The Internet Protocol (IP) assigns a unique address to each device, enabling communication over the internet.
 - Two versions
 - IPv4: 4 Byte long addresses
 - IPv6: 16 Byte long addresses and more flexibility with option fields
- IPv4
 - Dotted-decimal notation
 - Devices use bit representation

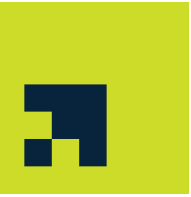


Exercise

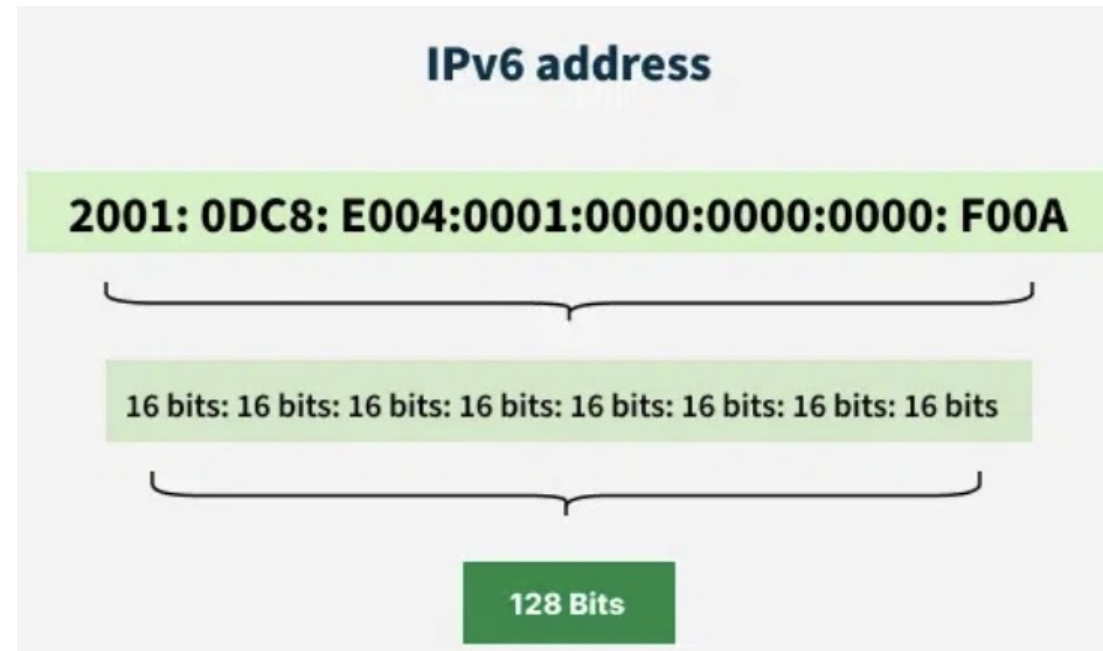


- Represent the following IP as a bit pattern: 9.152.255.3
- Which human-readable IP is the following bit pattern?
01101100 11101111 11000010 00111101

Exercise

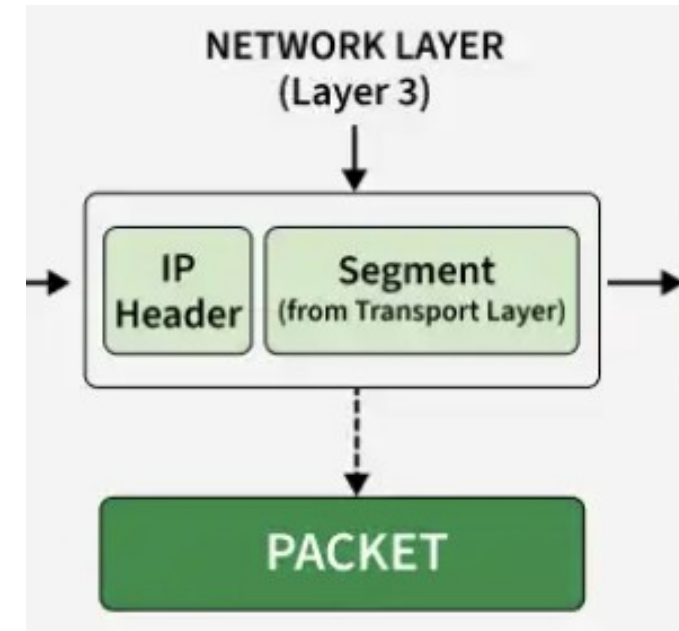


- An IPv6 address consists of eight groups of four hexadecimal digits separated by ':'
 - each Hex digit representing four bits so the total length of IPv6 is 128 bits.

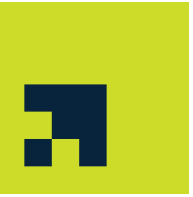


Packet

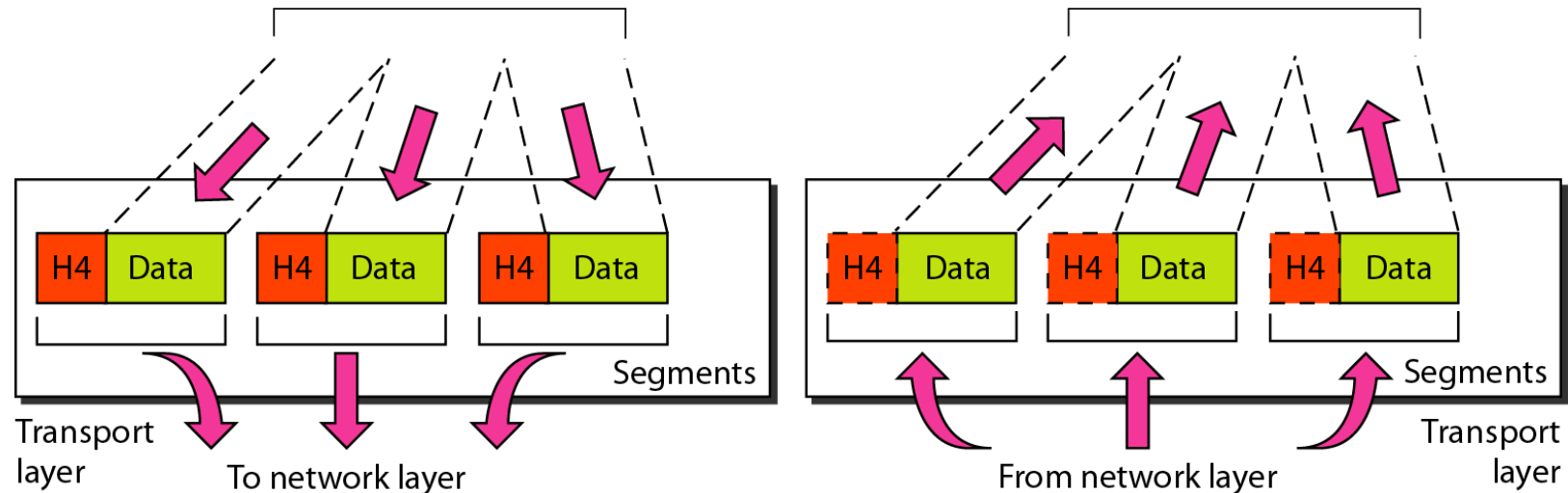
- An IP packet consists of a Header(typically 20 bytes, but can go up to 60 bytes) and Body (contains the segment from the transport layer).
- The Fields in the IP Header are:
 - Source IP Address: origin of the packet.
 - Destination IP Address: final destination of the packet.
 - TTL (Time To Live): Defines the packet's lifespan. It decreases by 1 at each hop (router). When TTL reaches 0, the packet is discarded to prevent infinite looping.
 - Identification: Helps in reassembling fragmented packets at the destination.
 - Protocol: Indicates the transport layer protocol used in the payload (e.g., TCP or UDP).
 - Version: Specifies the IP version being used (IPv4 or IPv6).
 - Flags and Fragment Offset: Used to manage fragmentation and reassembly of packets.



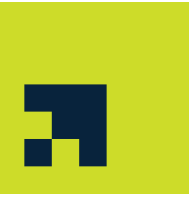
Transport layer



- The transport layer is responsible for the delivery of a message from one process to another.
- At the Transport layer addresses are called Port Address (16-bit addresses).
- TCP/IP model defines 3 protocols :
 - TCP (Transmission Control Protocol)
 - UDP (User Datagram Protocol)
 - SCTP(Stream Control Transmission Protocol)

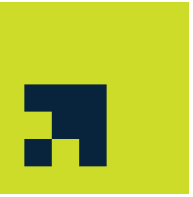


Protocols

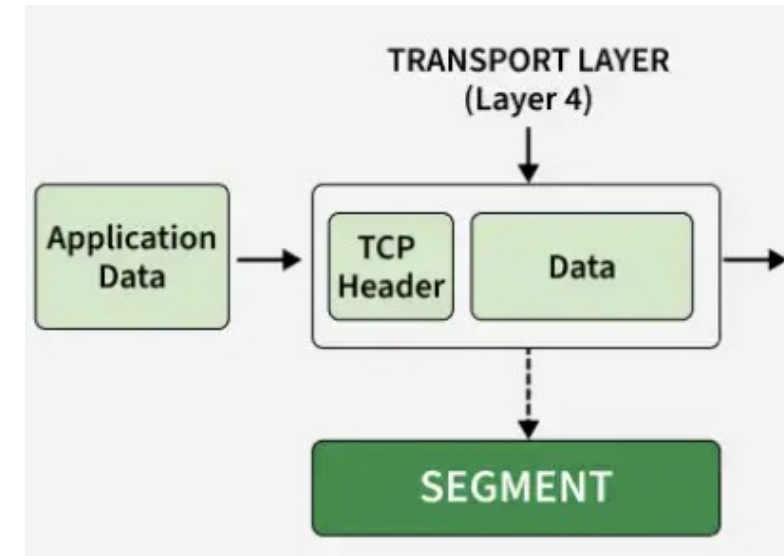


- User Datagram Protocol
 - Packet-oriented
 - Connection-less protocol
 - Unreliable transmission
 - Unicast, multicast, broadcast supported
- Transmission Control Protocol
 - Stream-oriented
 - Connection-based protocol
 - Reliable transmission
 - Only unicast

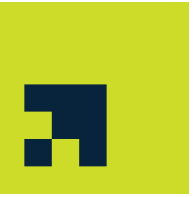
TCP Header



- The Key Components of a TCP Header are:
 - Source Port: Identifies the sending application or service.
 - Destination Port: Specifies the target application or service on the receiving device.
 - Sequence Number: Ensures that segments are reassembled in the correct order.
 - Flag Bits: Control communication (e.g., start/end of transmission, acknowledgment, or segmentation control).
 - Checksum: Verifies data integrity by detecting errors during transmission.
 - Options: Provides additional configurations such as segment size and performance tuning.



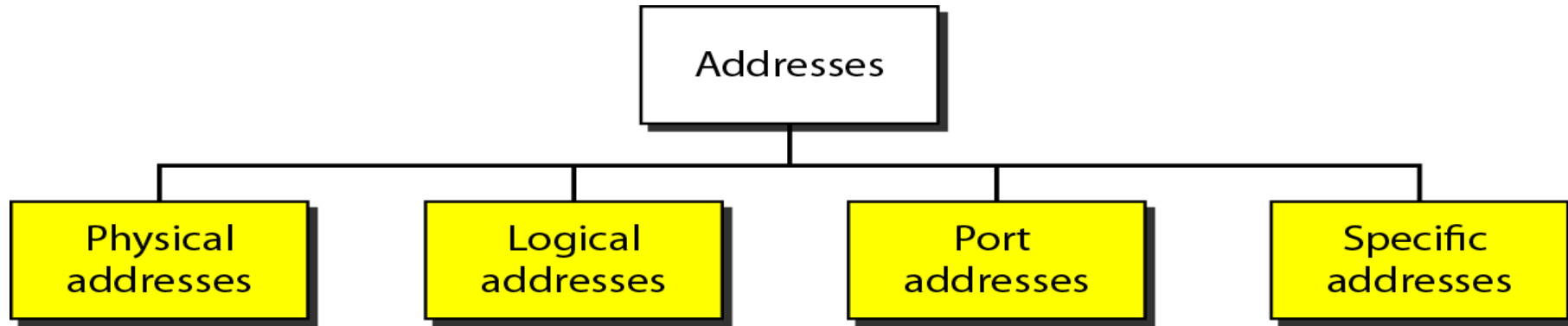
Application layer



- The application layer exchange messages between each other.
- TCP/IP model defines the following protocols:
 - FTP: File transfer protocol is used for transferring files from one host to another .
 - Telnet: A service that enables users on the internet to log onto remote systems from their own host system.
 - HTTP: Hyper text transfer protocol used for network file transfers in WWW environment
 - SMTP: Simple mail transfer protocol used to send electronic mail on the internet.

Addressing

Four levels of addresses are used in an internet employing the TCP/IP protocols



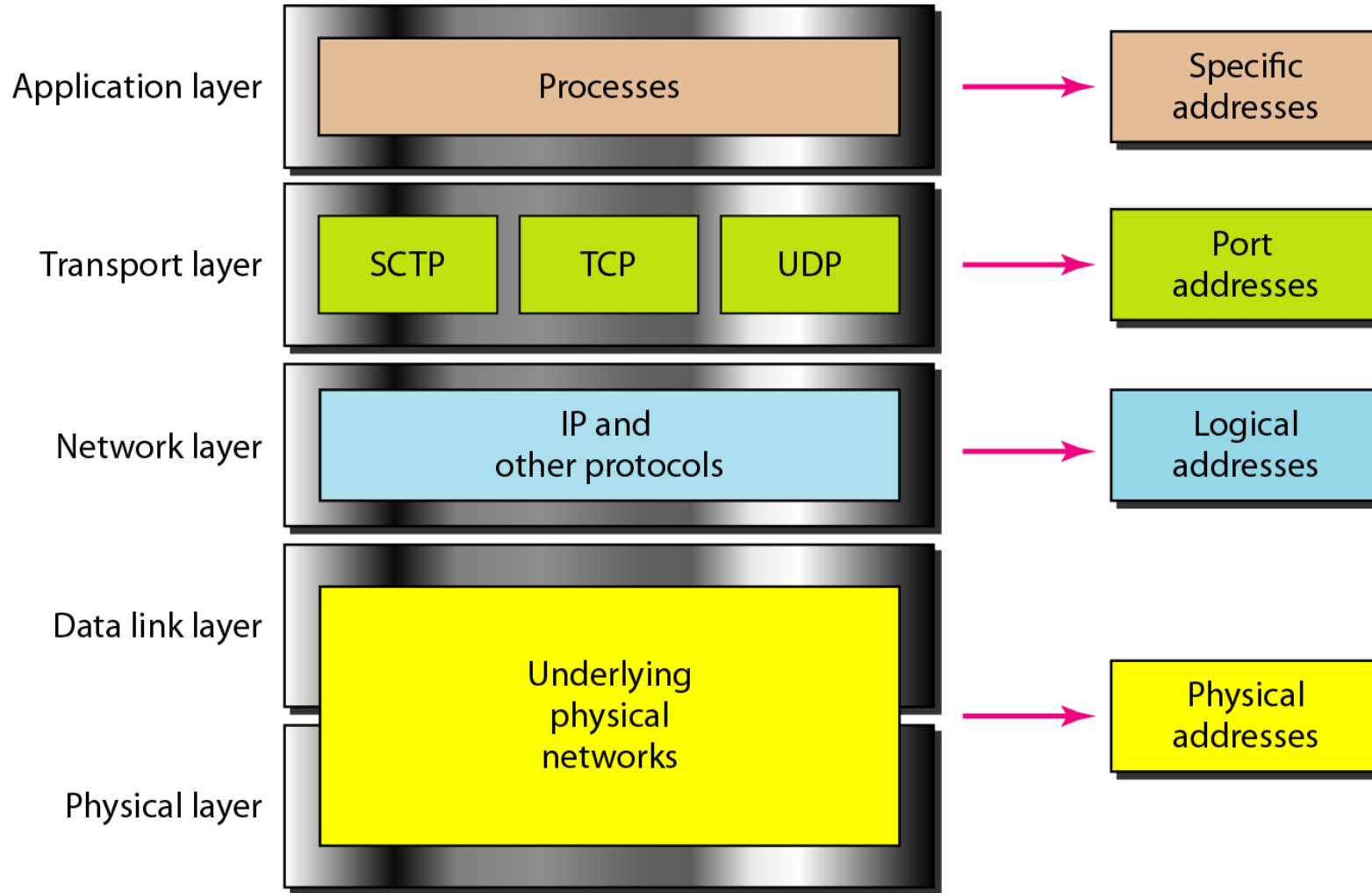
Examples for Specific addresses:

- **e-mail addresses** (*Lhajr@yahoo.com*) to define the recipient of an e-mail
- **URL addresses** (*www.ksu.com*) to find a document on the world wide web

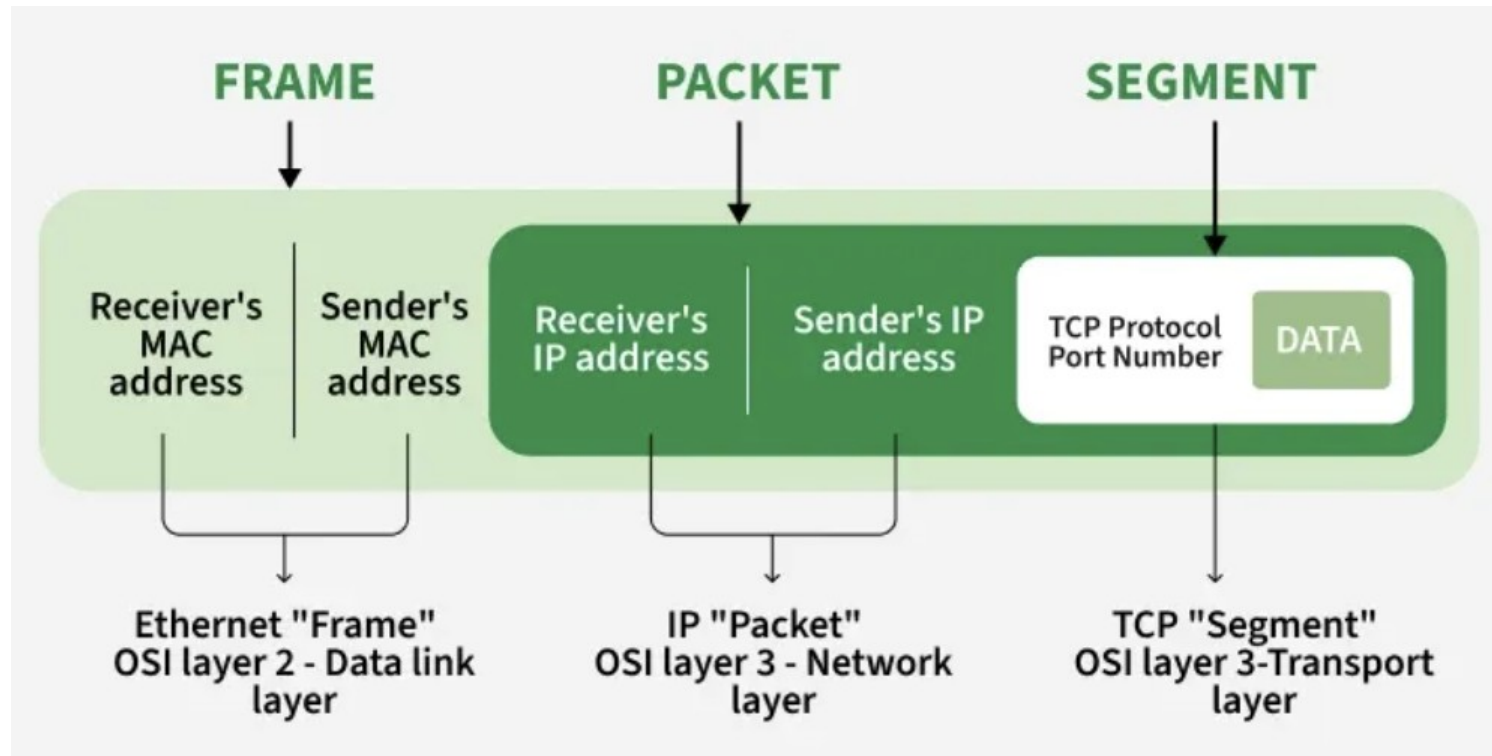
The addresses get changed to the corresponding port and logical addresses by the sending computer

The physical addresses will change from hop to hop, but the logical addresses remain the same.

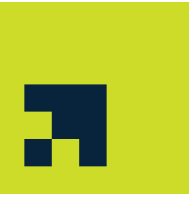
Relationship of layers and addresses in TCP/IP



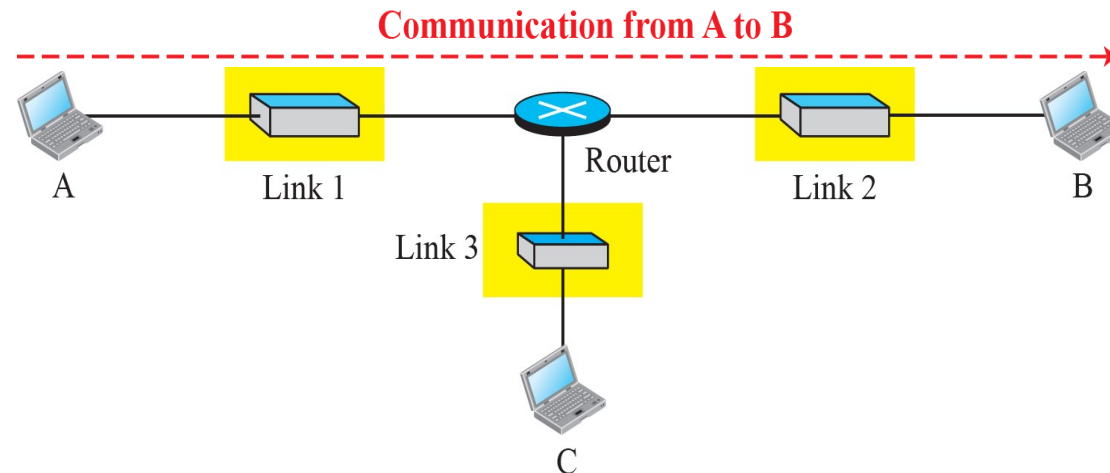
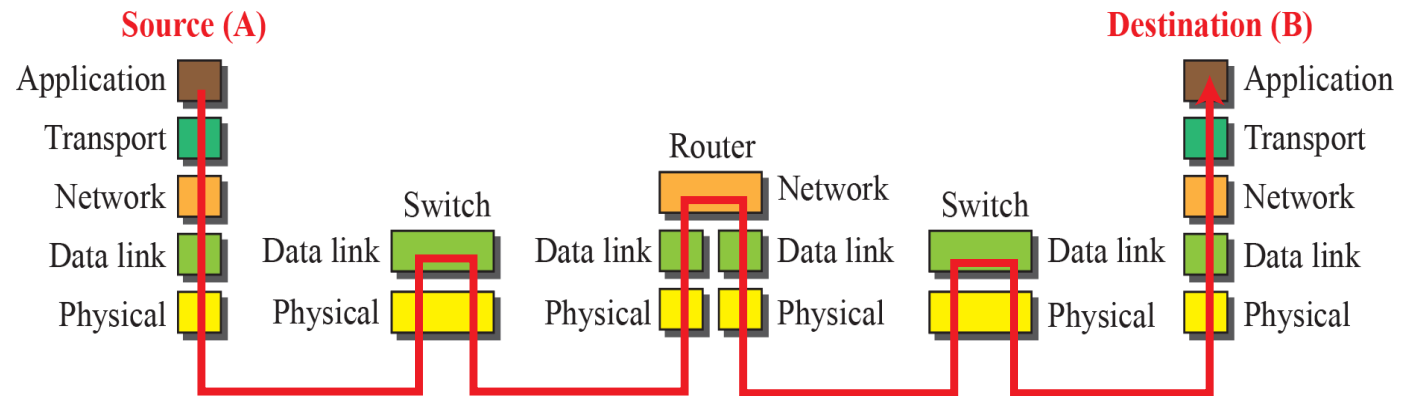
Structure of Segment, Packet, and Frame



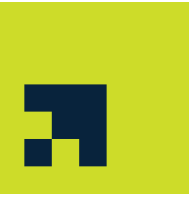
Communication through an internet



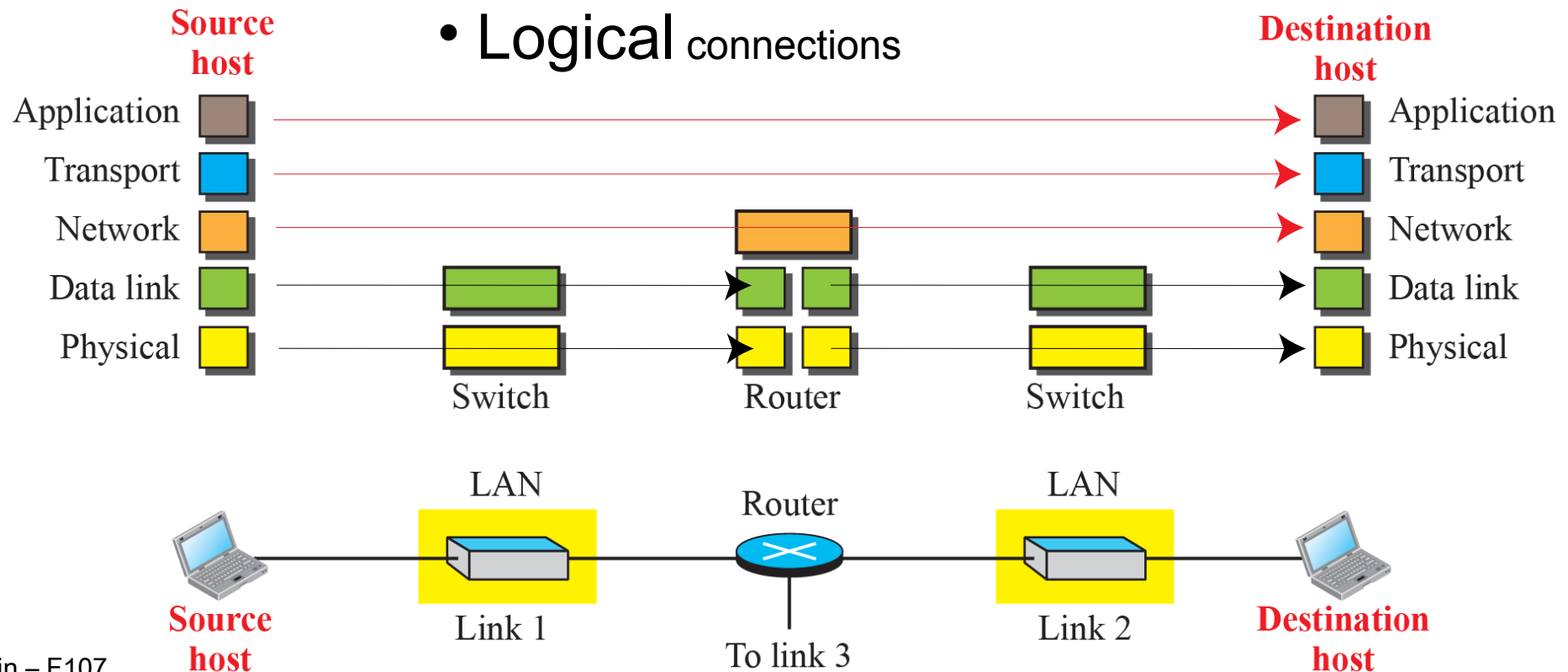
- We assume that we want to use the suite in a small internet made up of three LANs (links), each with a link-layer switch. We also assume that the links are connected by one router.



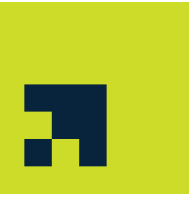
Layers in the TCP/IP Protocol Suite



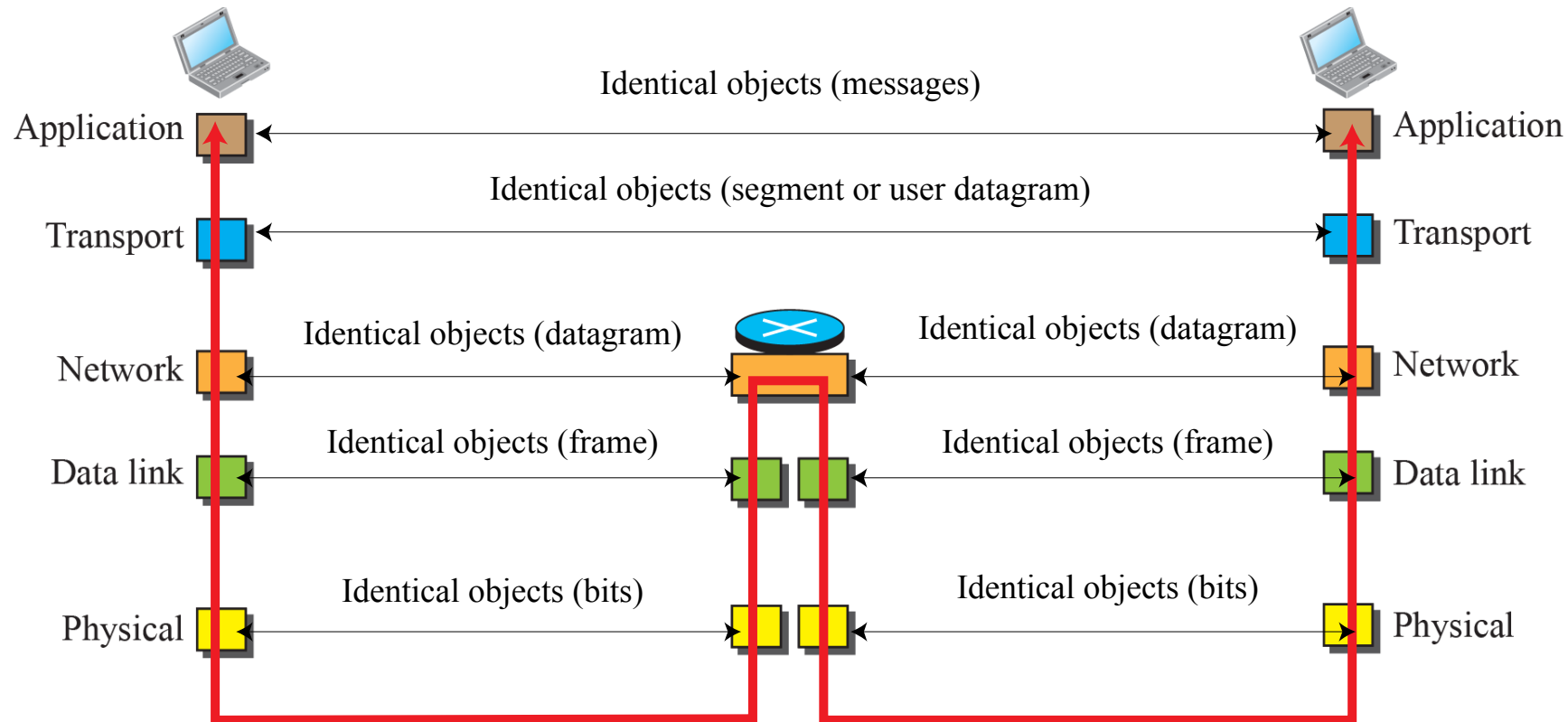
- To better understand the duties of each layer, we need to think about the logical connections between layers.



Identical objects in the TCP/IP protocol suite

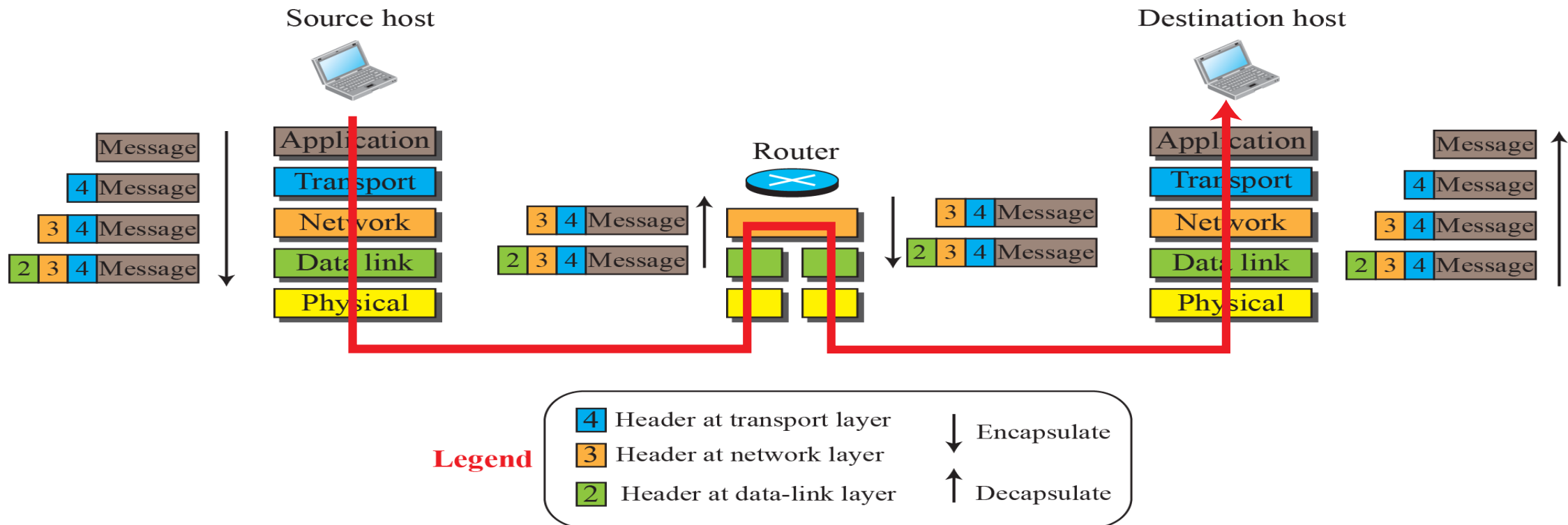


Notes: We have not shown switches because they don't change objects.

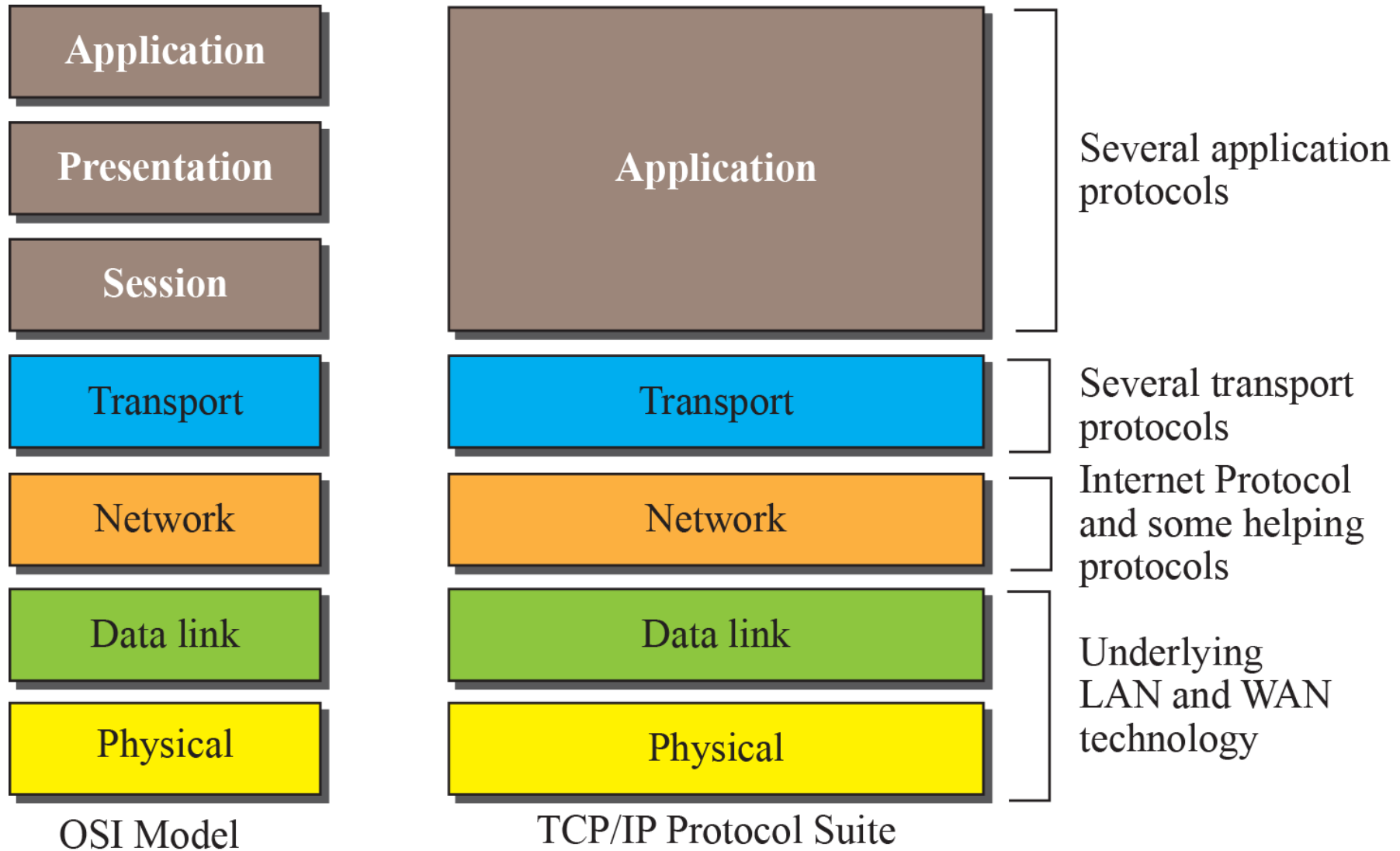


Encapsulation and Decapsulation

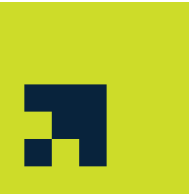
One of the important concepts in protocol layering in the Internet is encapsulation / decapsulation.



TCP/IP and OSI mod



Network Infrastructure Devices



ROUTER

Device that forwards data packets from one network to another. Based on address of the destination network in the incoming packet and an internal routing table, the router determines which port (line) to send out the packet.



SWITCH

Device that connects devices together on a computer network, by using packet switching to receive, process and forward data to the destination device.



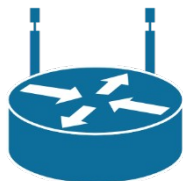
NAT Gateway

Device that connects two different networks, typically a public network and a private network. It allows multiple devices on the internal network to use one single external IP address. It translates between both networks, also considering the involved port numbers.



WIRELESS ACCESS POINT

Device that allows wireless devices to connect to a wired network using Wi-Fi. The AP usually connects to a router as a standalone device, but it can also be an integral component of the router itself.



BRIDGE

Device that connects multiple local networks, optionally even of different MAC mechanisms.

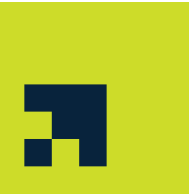


HOSTS

Host devices that actually run applications.



Device Identity



- How to identify a device?
 - Addresses or names

```
Wireless LAN adapter WiFi:

Connection-specific DNS Suffix  . : home
Description . . . . . : Intel(R) Dual Band Wireless-AC 8265
Physical Address. . . . . : 98-2C-BC-B7-3C-02
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
IPv4 Address. . . . . : 192.168.1.127(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : 30 October 2020 23:35:58
Lease Expires . . . . . : 05 November 2020 18:57:36
Default Gateway . . . . . : 192.168.1.254
DHCP Server . . . . . : 192.168.1.254
DNS Servers . . . . . : 192.168.1.254
NetBIOS over Tcpip. . . . . : Enabled
```

```
Windows IP Configuration

Host Name . . . . . : LAPTOP-M664ARP4
Primary Dns Suffix . . . . . :
```